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Mapping annual center-pivot irrigated cropland in Brazil during the 1985–2021 period with cloud platforms and deep learning

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ABSTRACT

As an important agricultural facility that supports efficient crop production and precise water management, center-pivot irrigated cropland (CPIC) plays a critical role in increasing crop yield and ensuring food security globally. Since the 1980s, CPIC has expanded significantly and has become one of the most valuable irrigated croplands in Brazil. However, national-scale and long-term CPIC products are currently unavailable, making spatiotemporal analysis for related ecological and economic factors challenging. In this study, we proposed a novel framework combining cloud platforms and deep learning algorithms to produce annual CPIC maps (ACMs) with 30 m spatial resolution in Brazil from 1985 to 2021. Leveraging all available Landsat-5, 7, and 8 images in Brazil, we generated three-band annual composite images to obtain stable image features for CPIC and alleviate the heterogeneity of CPIC. Then we constructed a large-scale training dataset covering different climatic zones of Brazil and spanning multiple years. Based on this dataset a CNN model was trained and applied to the whole of Brazil and generalized to other years. We deployed the CNN model to the cloud platform and combined it with Google Earth Engine to produce Brazil's annual CPIC Maps (ACMs). Moreover, we analyzed the spatiotemporal dynamics of CPIC and explored the impact of CPIC expansion on forests. The accuracy assessment demonstrated that our ACMs reached an average mIoU of 0.947 ± 0.021 for all study years, and the estimated CPIC area was highly consistent with official statistics. Our results indicated that Brazilian CPICs reached 1.8×10^6 ha in 2021, which was 45 times more than that in 1985. CPIC expansion occurred mainly in the states of Minas Gerais, Goiás, and Bahia over the past few decades, with an average annual increase in area of over 8,000 ha. We also found that CPIC expansion had encroached on 42,000 ha of shrubs and 25,000 ha of forests since 1985, and the impact was continuing and intensifying. To the best of our knowledge, the produced ACMs are the longest and most spatially detailed CPIC products in Brazil that reflect the CPIC spatiotemporal changes in the last four decades and can provide essential information services for Brazil's water resource management and agricultural decision-making.

1. Introduction

Center-pivot irrigated cropland (CPIC) is a critical component of irrigation and plays an essential role in improving water use efficiency and increasing food production (Chai et al. 2016; Kang et al. 2017; Rosa et al. 2020; Wang et al. 2021). Pumping groundwater and managing water resources, CPIC can provide timely and adequate irrigation to crops during their critical growing periods and contributes to high crop yields and resisting agricultural drought (Cooley and Smith 2022). With limited natural resources, such as land and freshwater (Rulli et al. 2013), developing CPIC to increase food production (FAO 2017) is thus one of the effective solutions to support the anticipated global population growth (McConnell and Viña 2018; Wang et al. 2021).

In Brazil, CPIC has become one of the most important and fastest-expanding irrigation types (ANA 2019; Multsch et al. 2020). A recent study showed that CPIC increased by 58,000 ha per year from 2014 to 2019, and its share of new irrigation expanded from 39 % in 2000 to 46 % in 2018 (ANA 2019). Increased CPIC mitigates seasonal climatic impacts and agricultural risks and ensures production and food security stabilization (de Albuquerque et al. 2020). In addition, the rapid CPIC

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expansion in the past decades has aroused public concerns about the environment, as it may lead to deforestation (Costa et al. 2007; Gibbs et al. 2010; Morton et al. 2006), groundwater resource overuse (Deines et al. 2017; Spera et al. 2016), and regional climate change (Silverio et al. 2015). To strike a balance between agricultural development and environmental conservation and formulate strategies toward sustainable development, knowing the expansion patterns and ecological impacts of CPIC over the last few decades is necessary. Therefore, producing Brazil's long-term annual CPIC maps is essential and urgent.

Remote sensing technology is a powerful tool for large-scale irrigation mapping and delineating its changes over time (Belgiu and Csillik 2018; Li et al. 2021; Xiong et al. 2017). The Brazilian National Water Agency (Agência Nacional de Águas, or ANA) produced eight Brazilian CPIC maps spanning from 1985 to 2019 at approximately five-year intervals (ANA 2019). However, since these CPIC products were generated through expert visual interpretation of remote sensing images, this approach would consume a large amount of manpower and material resources and its accuracy was largely dependent on the experiences of interpreters. The circular shape is the most distinctive characteristic for identifying CPIC from other irrigation types in satellite images (Saraiva et al. 2020). According to this characteristic, several studies have attempted to automatically extract CPIC directly from remote sensing images. For instance, Rodrigues et al. (2020) employed the Circular Hough Transform (Duda & Hart 1972) to detect circular targets from Landsat-8 image and subsequently used the time-series Moderate Resolution Imaging Spectroradiometer (MODIS) data to eliminate the circles that did not correspond to center pivots. Johansen et al. (2021) adopted an object-based approach to map the CPIC in Saudi Arabia from 2013 to 2015. However, these methods are only effective for easily distinguishable CPIC samples and are difficult to adapt to complex situations.

In recent years, convolutional neural network (CNN) methods have shown great potential in processing spatial patterns (e.g., edges, textures, and shapes) of images (LeCun et al. 2015) and have been applied to extract Brazilian CPIC from remote sensing imagery (de Albuquerque et al. 2020; Tang et al. 2021a). Tang et al. (2021a) employed the object detection network in conjunction with Circular Hough Transform to extract CPIC in central Brazil. Tang et al. (2021b) extracted the edge texture of images and integrated them into the original images to increase shape bias, and subsequently utilized the synthesized images to train a CNN model for CPIC extraction. However, these two-stage methods increased the complexity of the algorithm and were not suitable for large-scale applications. Some studies adopted the end-to-end approach to extract the CPIC from remote sensing images and achieved good performance in their respective study areas (de Carvalho et al. 2021; Saraiva et al. 2020). Several studies leveraged more image bands (de Albuquerque et al. 2020; de Carvalho et al. 2021) and multi-temporal images (de Albuquerque et al. 2021a; de Albuquerque et al. 2021b) as input features to train the CNN-based models to achieve higher prediction accuracy. These studies have demonstrated the effectiveness and advancement of CNN-based methods in extracting Brazilian CPIC. However, their study areas are limited to specific geographical extents, without considering the heterogeneity of CPIC and massive data processing costs faced by large-scale mapping. As a result, CPIC maps covering the whole of Brazil remain scarce.

The huge heterogeneity of CPIC across different regions of Brazil remains a challenge in building a generalized CNN model. The heterogeneity mainly comes from two aspects. First, multiple crops are often planted in a single CPIC, with each crop occupying a fan-shaped area, which leads to variations within the same CPIC. Second, CPIC in different climate zones has different planting patterns (e.g., single- and double-cropping systems) and phenological laws, which result in CPIC presenting different characteristics in the image. Although a previous study has attempted to identify the optimal season for distinguishing between CPIC and its background (de Albuquerque et al. 2020), determining a common time window across Brazil remains a challenge. Some studies used time-series images as input to avoid selecting the optimal periods (de Albuquerque et al. 2021a; de Albuquerque et al. 2021b), but these methods greatly increased data storage and computing costs, making them difficult to apply in large-scale and long-time series CPIC mapping. In addition, their study areas only cover a small portion of Brazil, and the constructed dataset cannot represent the CPIC characteristics of the entire Brazil. Therefore, selecting appropriate input features to alleviate the heterogeneity of CPIC and constructing a representative training dataset for Brazil remains an important problem that needs to be solved.

In the past, massive data and huge computational costs constrained long-term and nationwide CPIC mapping (Calderón-Loor et al. 2021; Gong et al. 2020; Yang & Huang 2021). Cloud computing platforms, such as Google Earth Engine (GEE) (Gorelick et al. 2017), drastically reduce the processing time of satellite images and expand the potential for long-term and large-scale mapping with high-resolution images (Hansen et al. 2013; Pekel et al. 2016; Xie et al. 2019a; Xie et al. 2019b; Xiong et al. 2017; Zhang et al. 2023). However, the GEE platform is inefficient in processing texture and shape features, and the provided classifier (Breiman 2001) cannot learn relationships among adjacent pixels, resulting in poor map accuracy and incomplete objects (Descals et al. 2021; Kattenborn et al. 2021). CNN-based methods overcome the deficiency in the mapping accuracy of classifiers provided in GEE. Therefore, combining the GEE platform and the CNN model becomes a potential solution to produce long-term, high-accuracy annual CPIC maps for Brazil.

This study developed a framework for producing the annual CPIC maps (ACMs) of Brazil for the period spanning from 1985 to 2021. We generated three-band annual composite images to obtain stable image features for CPIC and alleviate the heterogeneity of CPIC, and then constructed a large-scale training dataset covering different climatic zones of Brazil and spanning multiple years. Based on this dataset a CNN model was trained and applied to the whole of Brazil and generalized to other years. The developed framework is implemented on cloud platforms and can be easily extended to CPIC mapping for Brazil in future years. The developed framework is implemented on cloud platforms, enabling the processing of massive amounts of data and high-precision prediction of CPIC, resulting in the production of annual CPIC maps for Brazil. More importantly, this study analyzed the spatiotemporal dynamics of Brazilian CPIC over the last approximately four decades and explored the impact of its expansion on forests. Our study filled a knowledge gap in the development of Brazilian CPIC and provided essential information services for water resources management and agricultural decision-making in Brazil.

2. Study area and data

2.1. Study area

Brazil is the world's fifth-largest country by area, which covers about 8.5 million km^2, spanning nearly 39° (34.8°W–74.0°W) from east to west and 39° (5.3°N–33.8°S) from north to south, as illustrated in Fig. 1. The main climatic zones are the tropical, semi-arid, and subtropical zones (Alvares et al., 2013), accounting for 81 %, 5 %, and 14 % of the country, respectively. The northern part of Brazil is the Amazon basin, the largest rainforest in the world, which plays a critical role in preserving species diversity, regulating carbon cycle, and adjusting the global climate (Boulton et al. 2022; Cardoso et al. 2017). Most of the Amazon basin has an equatorial rainy climate with abundant precipitation. The southern part is the second largest plateau in the world, the Brazilian Highlands, which is dominated by a savannah climate. The southern end has many hills and plains, most of which have a humid subtropical climate.

The advantages of geography and climate make Brazil suitable for agricultural production and make it gradually develop into a global agricultural powerhouse (FAO 2020). Expanding irrigated areas is one

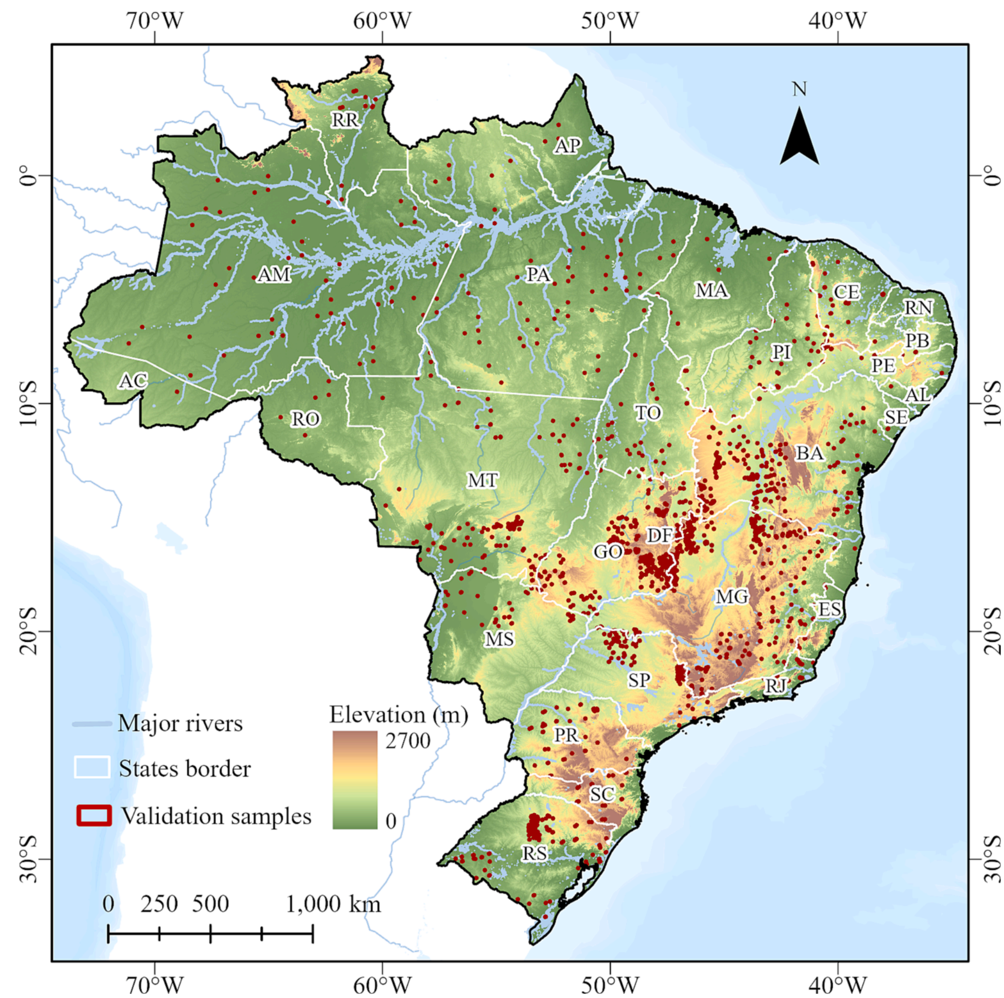

Fig. 1. Study area and distribution of validation samples.

of Brazil's strategies to increase agricultural production, among which developing CPIC is the primary measure. Brazil's CPIC has expanded rapidly over the past few decades, reaching approximately 1.56 million ha in 2019 (ANA 2019). CPIC is mainly used to plant soybeans, maize, cotton, wheat, and coffee in Brazil. CPIC can be divided into multiple sectors for planting different crops in the same period. Due to different climate zones, CPIC has different cropping patterns, such as single- and double-cropping systems. Hence, CPIC presents huge heterogeneity in different regions of Brazil.

2.2. Landsat imagery and preprocessing

Landsat is the only continuous, relatively high-resolution (30 m) dataset longer than 40 years, which allows us to produce long-term and historical CPIC maps for Brazil. The Landsat program is a series of Earth observation satellites managed by the United States Geological Survey and the National Aeronautics and Space Administration, which provides a continuous space-based record of Earth's land from 1972 to the present (Loveland and Dwyer 2012; Wulder et al. 2019; Wulder et al. 2022). Landsat 5–8 images have a temporal resolution of 16 days and a spatial resolution of 30 m. To produce annual CPIC maps (ACMs) for Brazil, we first collected and processed all available Landsat surface reflectance images in Brazil from 1985 to 2021, including the Landsat 5 Thematic Mapper (TM), Landsat 7 Enhanced Thematic Mapper Plus (ETM +), and Landsat 8 Operational Land Imager (OLI). All images are from GEE (https://earthengine.google.org) and have been conducted geometric and atmospheric correction, including cross-calibration among different sensors. For all Landsat images located in Brazil, those with cloud coverage greater than 80 % were discarded, and then the cloud and cloud shadow pixels were removed from each image using the CFMASK algorithm (Zhu & Woodcock 2014). After removing cloud-contaminated images, a total of 276,457 Landsat images were collected to produce ACMs in Brazil from 1985 to 2021.

Fig. 2 illustrated the number of Landsat images per year and the effective observation frequency in Brazil from 1985 to 2021. The number of images before 1999 was relatively small, with an average of 3,700 available scenes per year (Fig. 2a). With the launch of new Landsat satellites and the combined use of multi-satellite data, the number of available satellites each year has increased significantly. The high frequency of the effective observation of Landsat imagery in Brazil (Fig. 2b) makes it feasible for national-scale mapping applications.

2.3. Reference data collection

CPIC reference maps and statistic reports in the census years (i.e., 1985, 1990, 2000, 2005, 2010, 2014, 2017, and 2019) were obtained from the Brazilian ANA (https://metadados.snirh.gov.br/geonetwork/srv/por/catalog.search#/metadata/e2d38e3f-5e62-41ad-87ab-990490841073). They were used to evaluate ACMs in terms of area accuracy. A part of them was used to construct the training dataset for the U-Net model.

In addition, we collected independent validation samples to evaluate the accuracy of ACMs for each year. Specifically, we identified CPIC/Non-CPIC pixels whose categories remain constant over a period (e.g.,

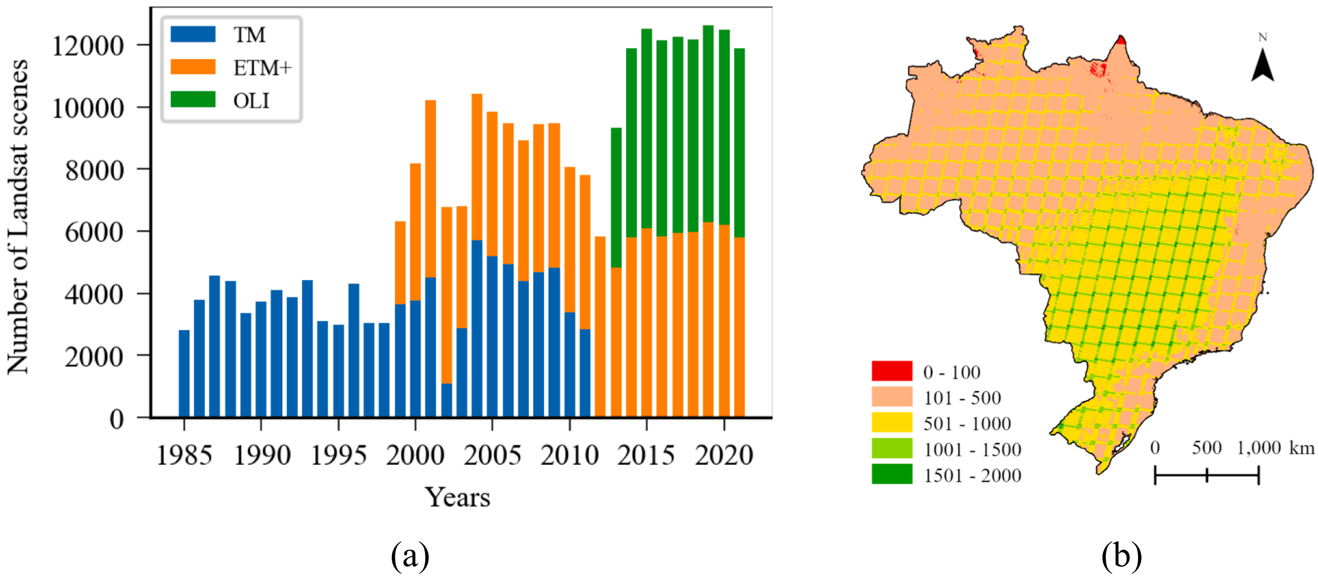

Fig. 2. Available Landsat images in Brazil from 1985 to 2021. (a) Number of Landsat images each year; (b) Effective observation frequency of Landsat images from 1985 to 2021.

five years) through visual interpretation to evaluate the accuracy of ACMs for all years in this period. Samples were validated by independent and experienced interpreters using the high-resolution (1 m) historical Google Earth imagery and Landsat Time Series Inspector. Validation samples were recorded as polygons, and all pixels within them were used for the pixel-level accuracy evaluation of ACMs. The 37 years from 1985 to 2021 were divided into six periods, and the samples collected in each period were used for the respective year-by-year accuracy evaluation of ACMs. Independent validation samples consisted of 2,641 polygons covering a total area of 200,000 ha. Details of validation samples are shown in Table 1.

We collected land-cover products (Zhang et al. 2021) with a 30 m resolution from 1985 to 2020 (five-year interval) to explore the encroachment of CPIC expansion on other land covers.

3. Methodology

We developed a mapping framework to automatically produce the 30 m ACMs of Brazil from 1985 to 2021. After describing the imagery and data preprocessing used in this research, we detailed the framework (Fig. 3) through the following steps: (1) selecting features and compositing images; (2) generating a training dataset; (3) training the U-Net classifier; (4) predicting on cloud platforms; (5) checking temporal consistency; and (6) evaluating ACMs.

3.1. Feature selection and image composition

CPICs present huge heterogeneity in Brazil due to the differences in crop types, cropping patterns, and phenological laws. The heterogeneity is present not only across different geographical regions but even within

Table 1
Number of validation polygons and their area for each validation year.

Validation year	Annual number of polygons		Total area (ha)
	CPIC	Non-CPIC	
1985–1989	61	76	11,036
1990–1999	240	268	39,654
2000–2004	239	243	36,624
2005–2009	244	243	37,868
2010–2013	246	259	39,400
2014–2021	257	265	38,763

adjacent CPIC plots. To mitigate image heterogeneity, we leveraged annual time-series Landsat images to obtain stable and reliable image features for CPIC. CPIC has significant vegetation reflection characteristics, bare soil characteristics, and moisture characteristics, which are determined by its nature as irrigated farmland. Therefore, related spectral indices, such as EVI (Huete et al. 1997), NDVI (Tucker 1979), BSI (Diek et al. 2017), NDMI (Gao 1996), and NDSMI (Yue et al. 2019), were considered as candidate features. Detailed information on these indices is shown in Table 2. EVI and NDVI reflect the vegetation growth status, and BSI captures the bare soil status from harvest to new planting. NDMI is sensitive to vegetation water content, and NDSMI indicates soil moisture after irrigation.

Although crop-growing seasons on CPIC vary significantly by crop type and climate type (Luo et al. 2020), they all go through key phenological periods of crop growth, such as the bare soil and vegetation periods. Therefore, features in these key periods can be obtained by the annual maximum composite of the above spectral indices. In addition, the annual median composite was applied for six Landsat surface reflectance bands to represent the average reflectance. Finally, 11 bands appeared as candidate features: $EVI_{\max}$, $NDVI_{\max}$, $BSI_{\max}$, $NDMI_{\max}$, $NDSMI_{\max}$, $blue_{median}$, $green_{median}$, red_{median}, nir_{median}, $swir1_{median}$, and $swir2_{median}$.

To speed up the optimization of deep learning algorithms and reduce data redundancy, only the three most important bands were retained in composite images. Therefore, obtaining the importance ranking of these features was necessary. Specifically, about 5,000 sample points both inside and outside the pivot were randomly selected to train a random forest model, which then provided the feature importance ranking of the 11 bands based on their information gain.

The average results of multiple experiments showed that $EVI_{\max}$,$BSI_{\max}$ and $green_{median}$ were the three most important features (Fig. 4). Therefore, we composited the new images using the three features and completed ACM production in Brazil based on these images.

Fig. 5 illustrates that our composite images can overcome the phenological differences of CPIC and achieve consistent CPIC features. Specifically, Site A and Site B are located in the Bahia and Rio Grande do Sul states, respectively, while Site C and Site D are adjacent CPIC parcels located in the Bahia state. The annual time series curves of their EVI values (Fig. 5a) and the false-color composite imagery from January 2020 (Fig. 5b), illustrated the significant heterogeneity of CPIC across different regions and within adjacent parcels. Fig. 5c illustrated that

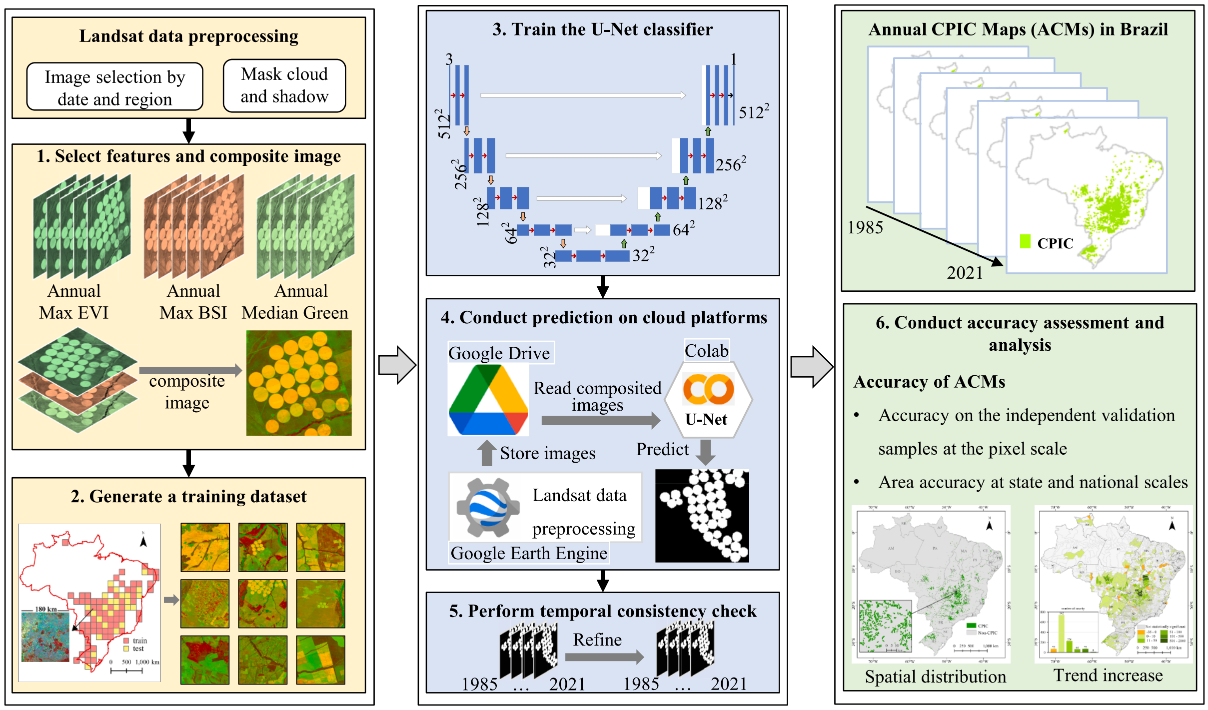

Fig. 3. Overview of the framework for producing the annual Center-Pivot Irrigated Cropland (CPIC) maps.

Table 2
Vegetation-, soil-, and moisture-related spectral indices. ρ_{blue}, ρ_{red}, ρ_{nir}, ρ_{swir1}, and ρ_{swir2} are surface reflectance of Landsat imagery at blue (450–520 nm), red (630–690 nm), near-infrared (760–900 nm), shortwave infrared 1 (1550–1750 nm), and shortwave infrared 2 (2080–2350 nm), respectively.

Name	Abbreviation	Equation
Enhanced Vegetation Index	EVI	$2.5 \times \frac{\rho_{nir} - \rho_{red}}{\rho_{nir} + 6\rho_{red} - 7.5\rho_{blue} + 1}$
Nominalized Difference Vegetation Index	NDVI	$\frac{\rho_{nir} - \rho_{red}}{\rho_{nir} + \rho_{red}}$
Bare Soil Index	BSI	$\frac{(\rho_{swir2} + \rho_{red}) - (\rho_{nir} + \rho_{blue})}{(\rho_{swir2} + \rho_{red}) + (\rho_{nir} + \rho_{blue})}$
Normalized Difference Moisture Index	NDMI	$\frac{\rho_{nir} - \rho_{swir1}}{\rho_{nir} + \rho_{swir1}}$
Normalized Difference Soil Moisture Index	NDSMI	$\frac{\rho_{swir1} - \rho_{swir2}}{\rho_{swir2} + \rho_{swir2}}$

CPICs with different phenology exhibited similar and consistent visual appearances in our composite images.

3.2. CPIC training dataset generation

Preparing representative training samples for CNN models, especially those that spatially cover the entire of Brazil and are consistent over time, is important. Considering the poor quality of ANA maps in earlier years, we collected CPIC maps for 2005, 2010, and 2017 from ANA reference data as ground truth labels. In particular, the training set was constructed using Landsat 5 and 7 images acquired in 2005 and 2010. Additionally, the training set also included Landsat 7 and 8 images from 2015. Therefore, the training dataset constructed from this has image characteristics of different Landsat sensors. To construct representative training samples and to select the optimal model during the training process, we divided these maps into training and testing regions according to their geographical locations. First, the entire Brazil area was divided into regular grids, each grid covering a region of approximately 180 × 180 km. Then, all grids were grouped into five density classes according to the number of CPICs in each grid. We adopted the stratified sampling strategy to divide the grids into training regions and test regions in a ratio of 7:3 according to the CPIC density class of each grid. The locations of the training and test regions are shown in Fig. 6a. Then, the images in the training regions were cut into patches with a size of 512 × 512 pixels, and were further divided into the training set and validation set in a ratio of 9:1. Finally, the number of patches for training, validation, and testing is about 18600, 2100, and 6800, respectively.

However, 80 % of the training samples did not contain CPIC targets. The imbalance of training samples would lead to fluctuations in the model training process and worse learning ability of the model (Li et al. 2018). Inspired by CutMix (Yun et al. 2019), we developed a data augmentation strategy named RandomMix that combines negative and positive samples to construct new samples, thereby increasing the proportion of CPIC pixels in the samples. Fig. 6b illustrates the specific steps for new sample generation. The implementation steps of the RandomMix are summarized in Algorithm 1.

Algorithm 1 RandomMix

Input: Dataset $D^{M+N} = \{D^M_{pos} \cup D^N_{neg}\}$, positive sample number M, negative sample number N**For** $\boldsymbol{k = 1, 2, 3, \ldots, N}$ **do**

1. Select negative sample I^k_{neg} from D^N_{neg} and randomly choose region p^k_{neg} covering 256 pixels × 256 pixels within I^k_{neg}.
2. Select positive sample I_{pos} from D^M_{pos} and take a quarter of I_{pos} as p_{pos}; perform 90°/180°/270° random rotation.
3. Paste p_{pos} into the location of p^k_{neg} to generate new image I^k_{new}.

End for Combine new dataset $D^N_{new} = \bigcup^N_{i=1} I^i_{new}$ and D^M_{pos} to form final dataset $D^{M+N}_{new} = D^N_{new} \cup D^M_{pos}$.**Output:** D^{M+N}_{new}.

3.3. U-Net classifier training

This study adopted the U-Net model (Ronneberger et al. 2015) as the classifier because of its excellent performance in many large-scale mapping works (Brandt et al. 2020; Kruitwagen et al. 2021; Sirko et al. 2021) and CPIC extraction work (de Albuquerque et al. 2020). The U-Net model can extract multilevel semantic features through a series of

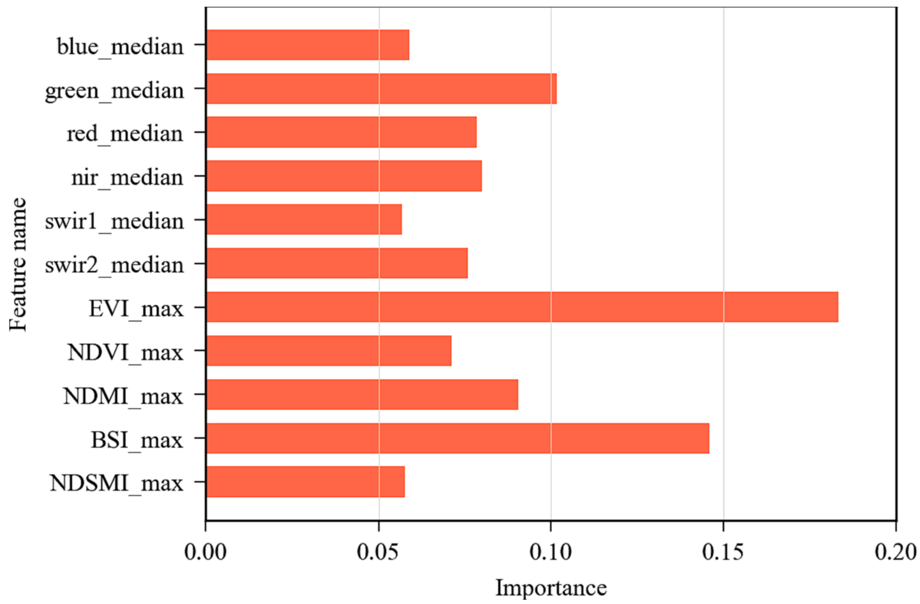

Fig. 4. Importance of the 11 candidate features.

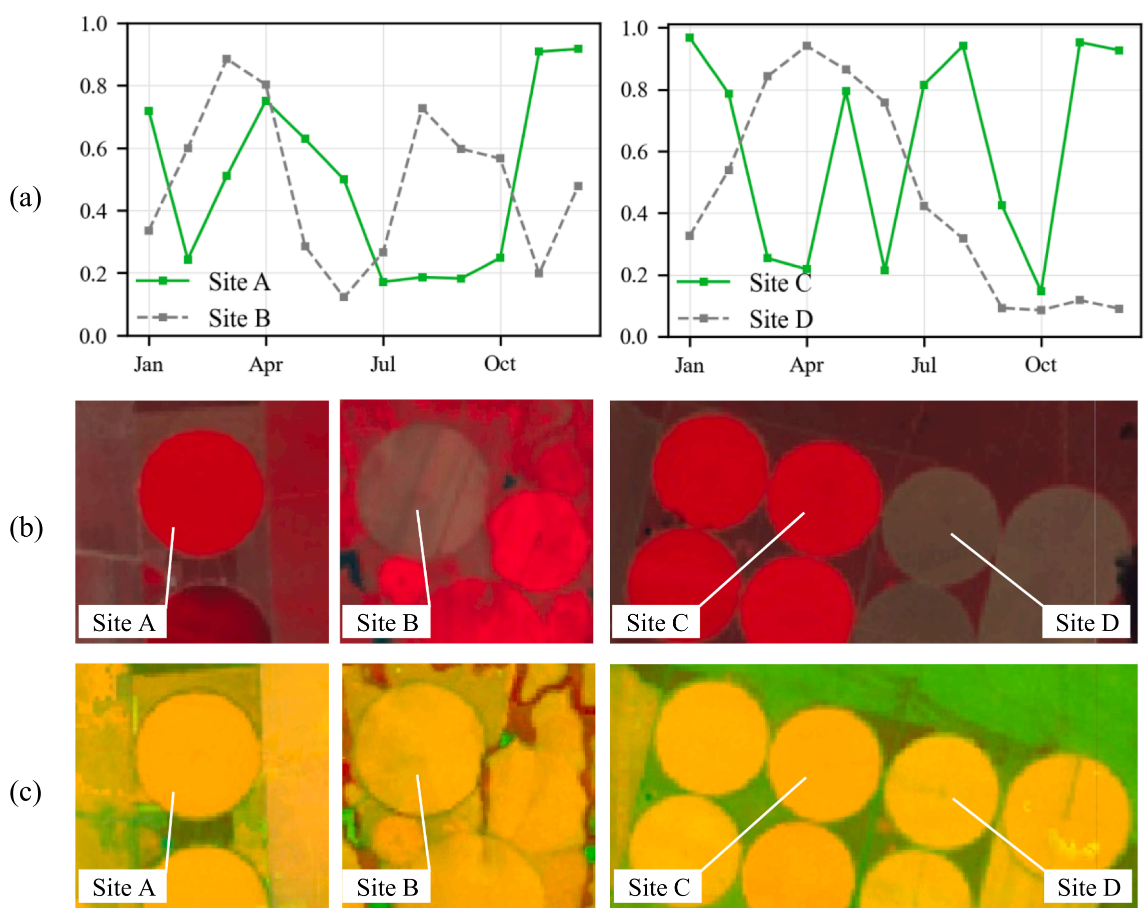

Fig. 5. The feature selection algorithm captured similar and consistent CPIC features with different phenological laws. (a) shows annual time-series curves of EVI, (b) is false-color composite imagery from January 2020. (c) illustrates similar and consistent visual appearances of CPIC in our composite images.

convolution operations and produce pixel-wise classification results.

The architecture of the U-Net is illustrated in Fig. 7. It consists of the encoder (left side) and decoder (right side), and receives 3-band images as input and outputs a single-band prediction map. The encoder follows the typical architecture of a convolutional network, which consists of repeated 3 × 3 convolutions, batch normalization (BN), and rectified linear unit (ReLU). After every two convolution operations, the max pooling is used to downsample the feature map once, and the channels of

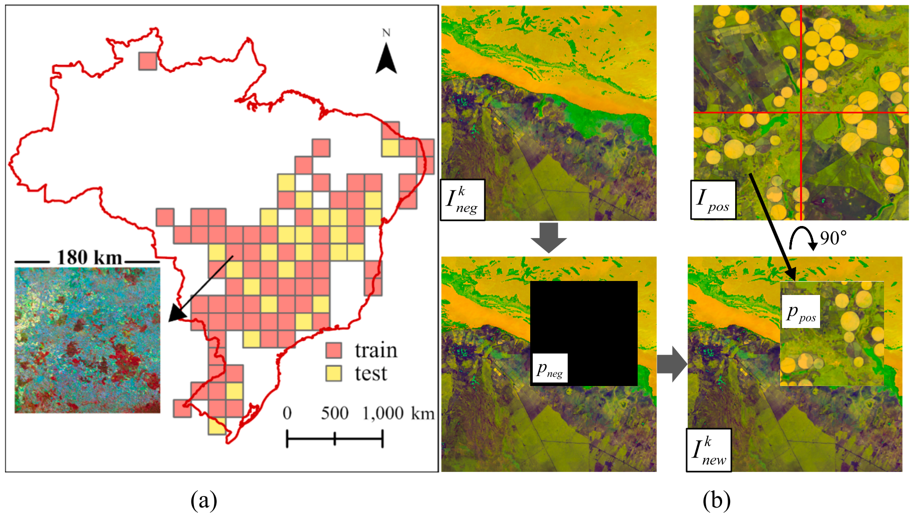

Fig. 6. Generation of training samples. (a) Spatial distribution of training and test regions. (b) Process for new sample generation: I_{pos} and I_{neg}^k represent positive and negative samples, respectively; p_{pos} and p_{neg} are the selected parts of them; I_{new}^k is the generated sample.

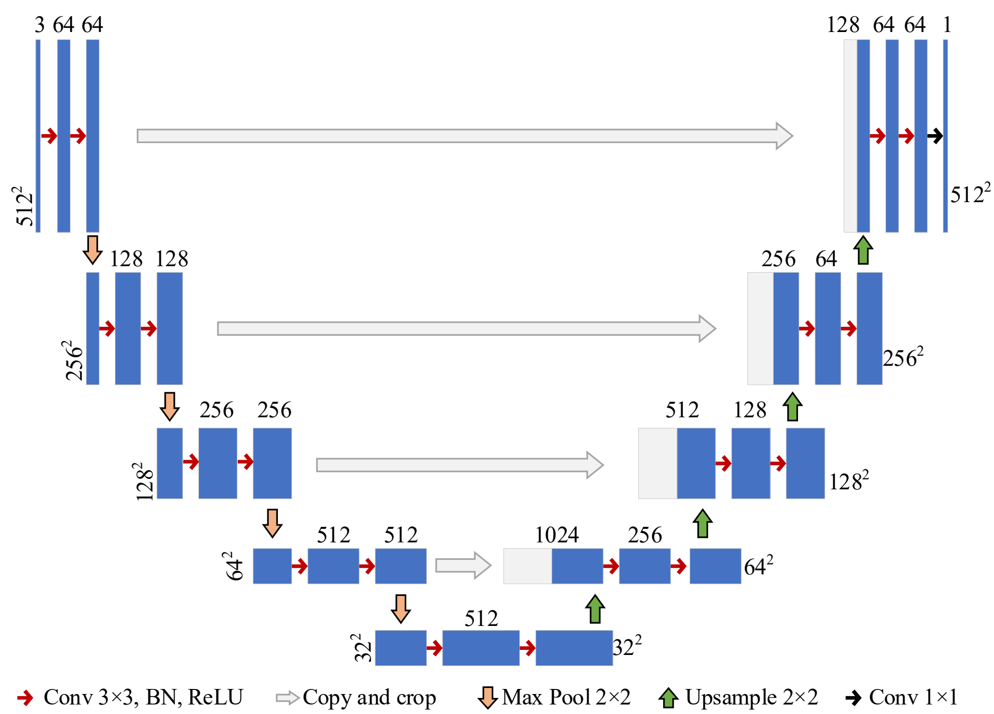

Fig. 7. The architecture of U-Net. Each blue rectangle corresponds to a multi-channel feature map. The numbers displayed to the left or right of the feature map indicate the size of them. The number above the feature map indicates the number of channels.

the feature map are doubled. In the decoder structure, the feature map is first upsampled by a factor of two and concatenated with the copied feature map from the encoder part along the channel dimension. After two convolution-BN and ReLU operations, a new feature map is obtained and used to recover a higher-resolution feature map. The final layer of the network uses a 1 × 1 convolution to map the 64-channel feature map to a prediction probability map.

A unified model is valuable for continuously monitoring CPIC and providing timely information to government decision-making. With the powerful feature representation ability of the CNN model, we aimed to train a unified U-Net model for Brazil and apply it to the CPIC mapping for all years. Specifically, the U-Net model was trained with generated training datasets, and the optimal model was determined by monitoring the Intersection over Union (IoU) metrics on the validation samples. Additionally, architectures such as HRNet (Sun et al. 2019) and U-Net with different backbones are compared for determining the optimal classification model. These backbones include ResNet (He et al. 2016) and Efficientnet (Tan and Le 2019).

The loss function is an important part of a CNN model, which is used to measure the difference between the predicted output and the actual output of the network. In our training datasets, the proportion of CPIC pixels in all images is only about 1.07 %, which causes a serious class imbalance problem. Tversky loss is a generalization of the Dice loss and Jaccard index, which can be used to control the trade-off between false positives and false negatives by adjusting different hyperparameters (Salehi et al. 2017). It is useful for segmenting small targets in unbalanced data of remote sensing images (Brandt et al. 2020; Kruitwagen et al. 2021). The superiority of the Tversky loss function on our dataset was shown in Table S1. Therefore, the Tversky loss was chosen as the loss function, and the RMSprop optimizer was applied with a learning rate of 0.0001 to minimize it.

The hyperparameters of the U-Net model in the training stage were set in the following manner. The U-Net was initialized with random weights, and the training epoch was set to 100. After the training stage, the optimum U-Net model was applied directly in the prediction of Brazilian CPIC from 1985 to 2021. Furthermore, the model can be directly applied to CPIC mapping in Brazil for future years without retraining.

3.4. Fast prediction on cloud platforms

To reduce time and storage costs in image transmission and to improve the mapping accuracy and speed, we deployed the CPIC mapping framework on cloud platforms, including GEE, Google Drive, and Colab (Carneiro et al. 2018). Landsat image preprocessing, annual maximum calculation, and image composition were conducted on the GEE platform, and the composited images were stored in Google Drive. The U-Net model was implemented on the Colab platform, which was responsible for reading composited images from Google Drive and generating CPIC prediction results. When predicting the whole image, we used a sliding window and set the overlap to 32 pixels to ensure a better reconstruction for the edge of the image.

3.5. Temporal consistency check

Although the annual composited images can ensure the acquisition of high-quality images, some regions still lack images under extreme cases due to cloud occlusion (Zhang et al. 2022). Missing images lead to CPIC misclassification in these regions. In practice, affected regions are not constant in different years. CPIC at the same location is also unlikely to disappear and appear again in a short period due to large investments. The above two prior knowledge allows us to correct affected regions using the results from adjacent years. Therefore, temporal filtering (Li et al. 2015) is used to correct the potential mistake classification of annual CPIC products. The equation for temporal filtering is as follows:

$$Prob_t = \frac{\sum_{q=t-T_w}^{q=t+T_w} con(L_q = L_t)}{1 + 2 \times T_w} \tag{1}$$

where $Prob_t$ is the probability of consistency; T_w denotes a temporal window, which was set to 1 in this study; t is the year from 1986 to 2020; $con(\ldots)$ means to check the consistency of the classification map in the year of t and q; L represents the pixel value of the classification map, of which the CPIC pixel is 1 and others are 0.

The probability of consistency of the current year with previous and after years within temporal window T_w was used to correct erroneous pixels. If the probability value is less than 0.5, then the current pixel value becomes its logical inversion, and vice versa.

3.6. Accuracy assessment

The ACMs conducted evaluations from three aspects: visual comparison with other mapping results (de Carvalho et al. 2021), quantitative accuracy assessment on an independent validation set, and comparison with statistical areas from ANA survey data (ANA 2019). The intersection over union (IoU), precision, recall, and F1-score were used to assess the accuracy of ACMs. We applied an error-adjusted area estimator (Olofsson et al. 2013) to obtain the mapped areas and the area uncertainty in the 95 % confidence interval (CI). In addition, country- and state-level CPIC areas aggregated from ACMs were compared to ANA statistics data for the census years.

4. Results

4.1. Comparison between CNN architectures

Fig. 8 illustrates the segment results and errors of different models on the test dataset. The results showed that all models achieved good prediction results for easily identifiable samples, as shown in Fig. 8(I, II). However, when dealing with difficult CPIC samples that were similar to the background, other comparison models exhibited varying degrees of missed detection errors, as illustrated in Fig. 8(III, IV). The U-Net model consistently achieved high accuracy across all test cases.

The quantitative comparisons of different models were shown in Table 3. The results showed that among all compared models, the normal U-Net architecture achieved the highest prediction accuracy, with mIoU and F1-score reaching 0.903 and 0.893, respectively. While the prediction results of models with advanced backbones have declined in mIoU, Recall, and F1-score metrics.

Therefore, the qualitative and quantitative analysis demonstrated that for the Brazilian CPIC segmentation task, the U-Net model outperformed all other models compared in the study. This conclusion is also consistent with the previous study (de Albuquerque et al. 2020).

4.2. Evaluation of the accuracy of the ACMs

de Carvalho et al. (2021) took the instance segmentation models to automatically extract the CPIC with the Landsat-8 imagery. Although they did not release their dataset and predicted maps, we still located the same geographical location based on the pictures in their paper and visually compared our results with theirs, as shown in Fig. 9. The compared result showed that our prediction maps were generally consistent with theirs, both achieving high accuracy. The results also illustrated that some CPICs were missed in their results, while our method was able to correctly detect these CPICs.

ACM accuracy was evaluated with independent validation samples at the pixel scale. The precision, recall, mIoU, and F1-score of CPIC products for all mapping years are shown in Fig. 10.

Our products achieved high accuracy on these metrics. Specifically, the average precision retained high values (0.996 ± 0.005) for all mapping years, which indicated that almost no commission errors existed in our CPIC products. The average value of recall across all mapping years was 0.948 ± 0.021, suggesting that the omission error (or underestimate) of our CPIC products was low. Moreover, the average mIoU and F1-score of ACMs reached 0.947 ± 0.021 and 0.971 ± 0.012, demonstrating that our CPIC products provided a high-quality characterization of CPIC distribution from 1985 to 2021.

In addition to pixel-based accuracy assessments, the state- and national-level aggregated CPIC areas between our products and ANA reports were compared. Fig. 11 compared the estimated CPIC area and the reported area from ANA in 27 states for eight census years. The results illustrated that our products provided accurate estimations of CPIC areas, with an R^2 of over 0.99 for most years.

The national-level CPIC area comparisons between our products and ANA reports (Fig. 12) also indicated that estimated CPIC areas were close to the official data. In the most recent census year (2019), the estimated CPIC area of Brazil was 1,578,280 ha, which was very close to the 1,564,555 ha reported by the ANA. Therefore, the accuracy evaluation shows the high quality and reliability of our ACMs and indicates that our methodology performs well for CPIC mapping in other years.

4.3. Spatial distribution of CPICs

Our framework generated 37 annual CPIC maps for Brazil. We presented the spatial distribution of CPICs in Brazil and their local details in 2021 (Fig. 13). The spatial distribution of CPICs in other years was shown in Fig. S1. At local details, our products captured field-level CPIC details and provided clear CPIC and non-CPIC boundaries. The

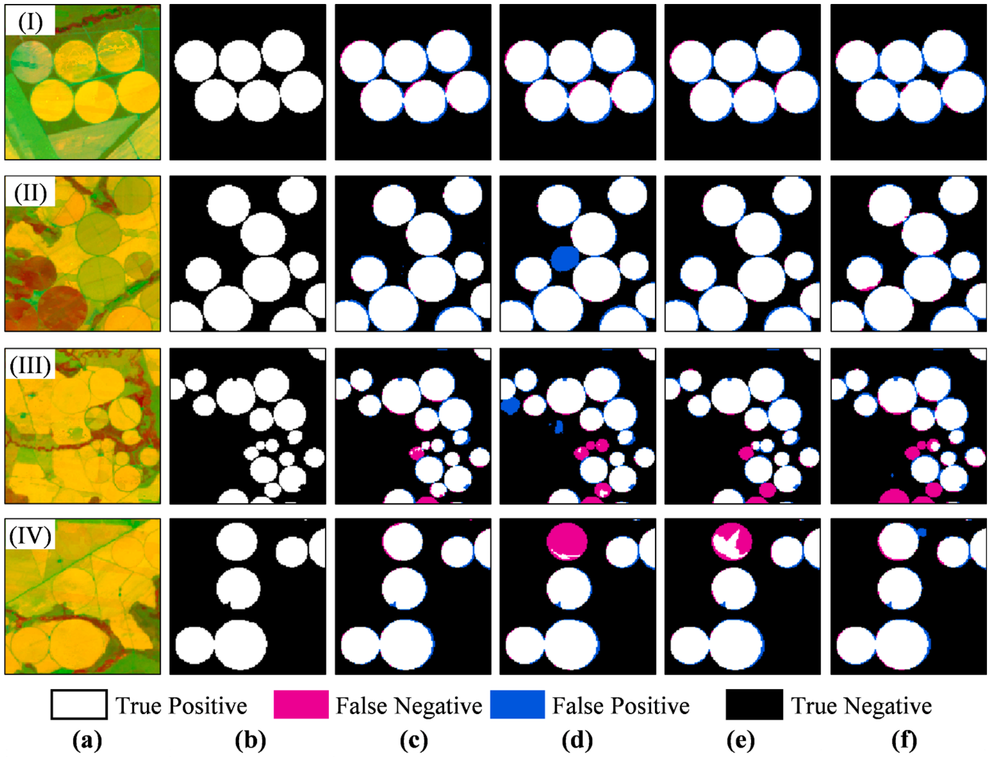

Fig. 8. Segment results and errors of different models on the test dataset. (a) Images of the test dataset. (b) Ground truth of CPIC. (c) U-Net. (d) ResNet101-UNet. (e) EfficientNetB5-UNet. (f) HRNet.

Table 3
Quantitative comparison of different models on the test dataset.

Model	Backbone	mIoU	Precision	Recall	F1-score	OA
U-Net	–	**0.903**	**0.898**	**0.888**	**0.893**	0.999
U-Net	ResNet50	0.887	0.879	0.866	0.873	0.999
U-Net	ResNet101	0.883	0.885	0.852	0.868	0.999
U-Net	EfficientNetB4	0.876	0.865	0.852	0.859	0.999
U-Net	EfficientNetB5	0.873	0.895	0.818	0.855	0.999
HRNet	–	0.879	0.857	0.868	0.863	0.999

nationwide map reflected the spatial distribution of CPICs in Brazil. CPICs were mainly distributed in the central and southern parts of Brazil. Specifically, CPICs were mainly distributed in six Brazilian states, including Bahia, Goiás, Mato Grosso, Minas Gerais, Rio Grande do Sul, and São Paulo, and their CPIC areas were all greater than 100,000 ha in 2021.

Table 4 shows the CPIC areas and percentages in all states of Brazil in 2021. Brazil's CPIC area in 2021 was about 1,809,530 ± 2,294 ha (with 95 % CI). The CPIC area in the six most distributed states accounted for more than 92 % of the total area. Specifically, Minas Gerais had the most CPIC area and contributed 553,452 ha, accounting for 30.6 % of the total area. By contrast, the states in the Amazon basin have few CPIC areas (less than 100 ha), such as Acre, Amapá, and Amazonas.

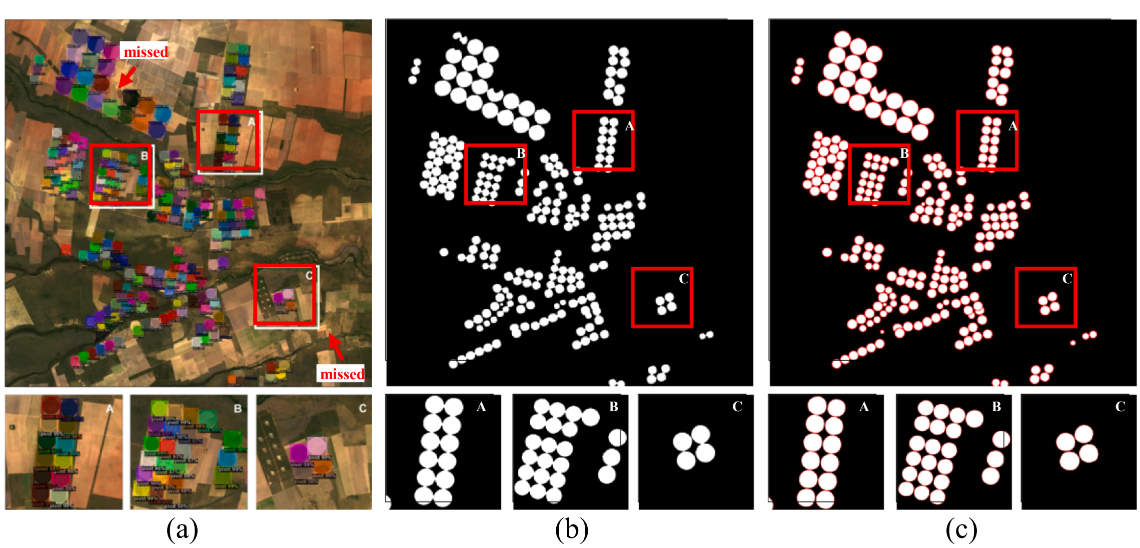

Fig. 9. Comparison of our products with existing products. (a) De Carvalho's CPIC map. (b) Our CPIC map. (c) Ground truth.

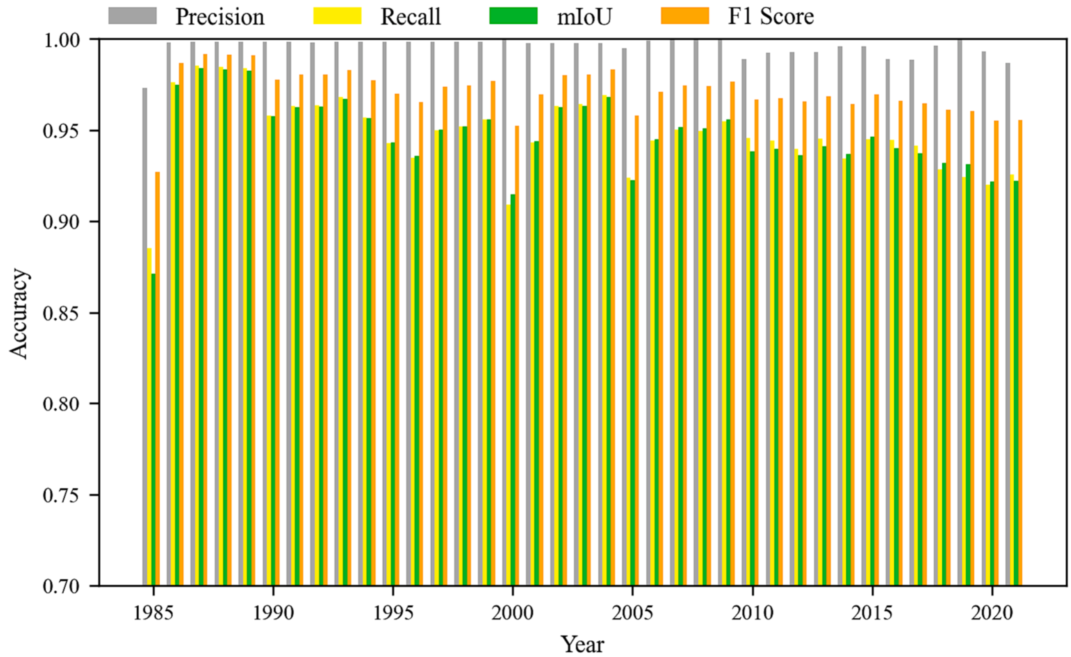

Fig. 10. The precision, recall, mIoU, and F1-score of CPIC products for each mapping year.

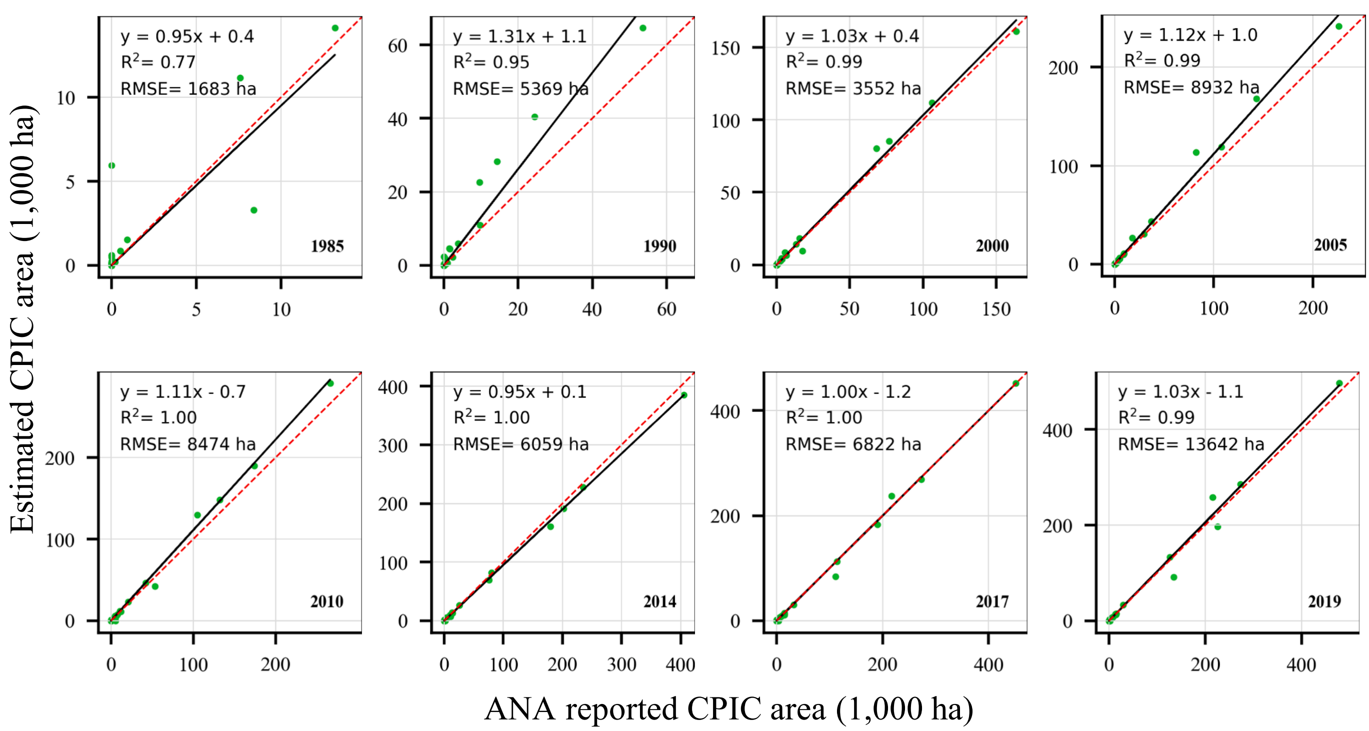

Fig. 11. State-level comparisons between our estimated and ANA-reported CPIC areas for eight census years. The dashed red line is the 1:1 line. Note that all linear regressions are statistically significant with $p < 0.001$.

4.4. CPIC area dynamics from 1985 to 2021

ACMs revealed the development trends of CPIC areas in Brazil from 1985 to 2021, as shown in Fig. 14. The CPIC area in each year indicated that Brazil had seen a sharp CPIC expansion from 1985 to 2021, increasing by 45 times from 39,857 in 1985 to 1,809,530 ha in 2021 (Fig. 14a). Moreover, the annual increased area (Fig. 14b) and increased rate (Fig. 14c) of CPIC varied in different years and presented different growth patterns.

We divided the CPIC growth pattern in Brazil into four periods on the basis of the annual increased area and rate. The first period spanned from 1985 to 1995, when the annual increased CPIC area stabilized at 3 $\times$ 10^4 ha with a high increase rate (more than 10 %). The second period spanned from 1996 to 2003, during which the annual increased area gradually increased and continued to peak in 2003. The third period spanned from 2004 to 2013, during which it presented a “V-shape” increase pattern. Specifically, the annual increased area gradually decreased until a trough was reached in 2007, and then it gradually

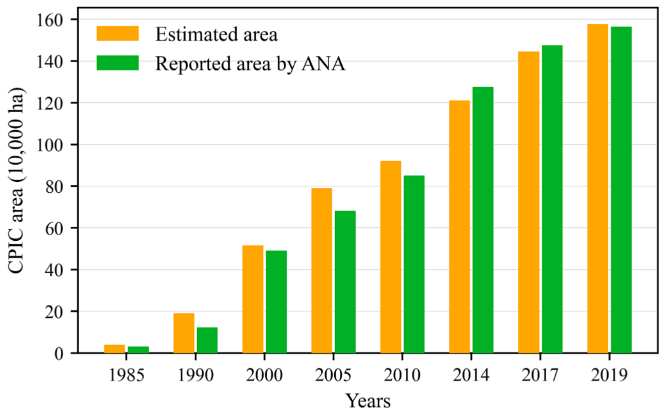

Fig. 12. Country-level comparisons between our estimated and ANA-reported CPIC areas for eight census years.

increased and continued to peak in 2013. The fourth period spanned from 2014 to 2021, during which it showed a new round of the "V-shape" pattern again and reached its peak in 2021. In addition, CPIC areas grew at different rates in different Brazilian states. Fig. 15 illustrated the inter-annual variations in the top 5 states with the largest CPIC area expansions from 1985 to 2021. Notable increases in CPIC areas were observed in Bahia, Goiás, Mato Grosso, Minas Gerais, and São Paulo, where their CPIC areas in 2021 were 90, 210, 183, 32, and 20 times larger than those in 1985, respectively (Table S2). The average annual increase area of CPIC in Bahia, Goiás, and Minas Gerais exceeded 8,000 ha per year, of which Minas Gerais' annual increased area was most significant and reached about 15,000 ha per year.

To further explore the growing CPIC trend from 1985 to 2021 at a fine scale, the trend analysis method was adopted to analyze the changing trend of the CPIC area in each county. Specifically, Sen's estimator (Sen 1968) was used to characterize the magnitude of each CPIC area change in each county, and the statistical significance of the changing trend was analyzed through Mann–Kendall (MK) test (Kendall 1946; Mann 1945). These counties with statistical significance ($p < 0.05$) illustrated the change rates in different regions and revealed more concentrated changes.

Fig. 16 illustrates the Sen's slope in each county and the number of counties with different slopes. Overall, 1,185 counties experienced significant changes, with 1,113 showing an increasing trend and 72 showing a declining trend. Among counties with significant trends, 742 counties showed a slightly increasing trend (Sen's slope is between 0 and 10), accounting for more than 60 %. The most rapidly expanding CPIC regions were distributed in nearly the Distrito Federal, east of the Bahia, and the center of Mato Grosso, where their Sen's slopes were greater than 500. Conversely, the CPIC in the north of the Minas Gerais and southwest of Pernambuco had declined over the past years, especially around the Sobradinho reservoir.

Furthermore, the pixel-level investigation (Fig. 17) illustrated the ability of our CPIC maps to capture pixel-level CPIC changes over time, such as sharp CPIC expansion in the east of the Bahia (site A), CPIC abandonment in the southwest of Pernambuco (site B), and the transfer change in the north of São Paulo (site C).

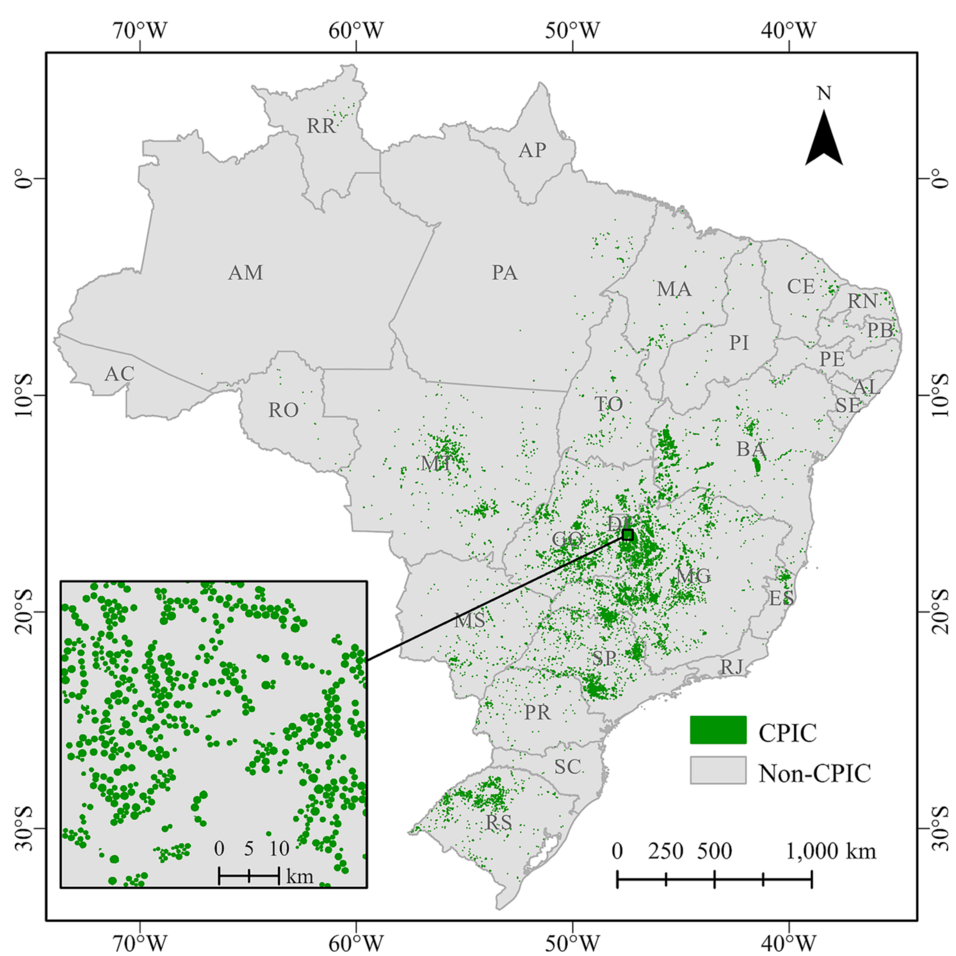

Fig. 13. Spatial distribution of nationwide CPIC areas and their local details maps in 2021. CPIC pixels are indicated by the green color.

Table 4
CPIC areas and proportions in different states of Brazil in 2021.

State	CPIC Area (ha)	Proportion (%)
Acre (AC)	0	0.0
Alagoas (AL)	567	0.0
Amapá (AP)	0	0.0
Amazonas (AM)	32	0.0
Bahia (BA)	297,180	16.4
Ceará (CE)	8,758	0.5
Distrito Federal (DF)	15,271	0.8
Espírito Santo (ES)	13,030	0.7
Goiás (GO)	317,259	17.5
Maranhão (MA)	10,960	0.6
Mato Grosso do Sul (MS)	47,658	2.6
Mato Grosso (MT)	155,114	8.6
Minas Gerais (MG)	553,452	30.6
Pará (PA)	7,610	0.4
Paraíba (PB)	1,280	0.1
Paraná (PR)	17,139	0.9
Pernambuco (PE)	696	0.0
Piauí (PI)	3,309	0.2
Rio de Janeiro (RJ)	77	0.0
Rio Grande do Norte (RN)	3,613	0.2
Rio Grande do Sul (RS)	112,090	6.2
Rondônia (RO)	604	0.0
Roraima (RR)	2,182	0.1
São Paulo (SP)	224,894	12.4
Santa Catarina (SC)	667	0.0
Sergipe (SE)	564	0.0
Tocantins (TO)	15,524	0.9
Total	**1,809,530**	**100.0**

5. Discussion and analysis

5.1. The benefits of our approach

Images at the beginning of the dry season may be beneficial for distinguishing the CPIC from the background (de Albuquerque et al. 2020). However, as Brazil spans nearly 39 latitudes and covers multiple climate zones, there is significant heterogeneity in the CPIC across different regions. While some regions may have already entered the dry season, others may still be in the vegetation period. Additionally, images obtained in the dry season may be subject to severe cloud pollution. Therefore, it is impractical to determine a common period as the optimal season for mapping the CPIC across all of Brazil. Previous studies took time-series images to avoid the trouble of selecting the optimal periods for different regions and to alleviate cloud pollution (de Albuquerque et al. 2021a; de Albuquerque et al. 2021b). But these methods greatly increased data storage and computing costs, making them difficult to apply in large-scale and long-time series CPIC mapping. Our study leveraged the time-series Landsat images throughout the year to construct three-band image features, which not only solves the problems of CPIC heterogeneity and cloud pollution but also alleviated the data storage and computing costs.

We used annual time series of Landsat images to calculate the annual maximum values of EVI, BSI, and the median of the green band, and combined them into a 3-band image to complete the annual CPIC mapping for Brazil. We compared the performance of different models and discussed the impact of input features on segmentation results. For

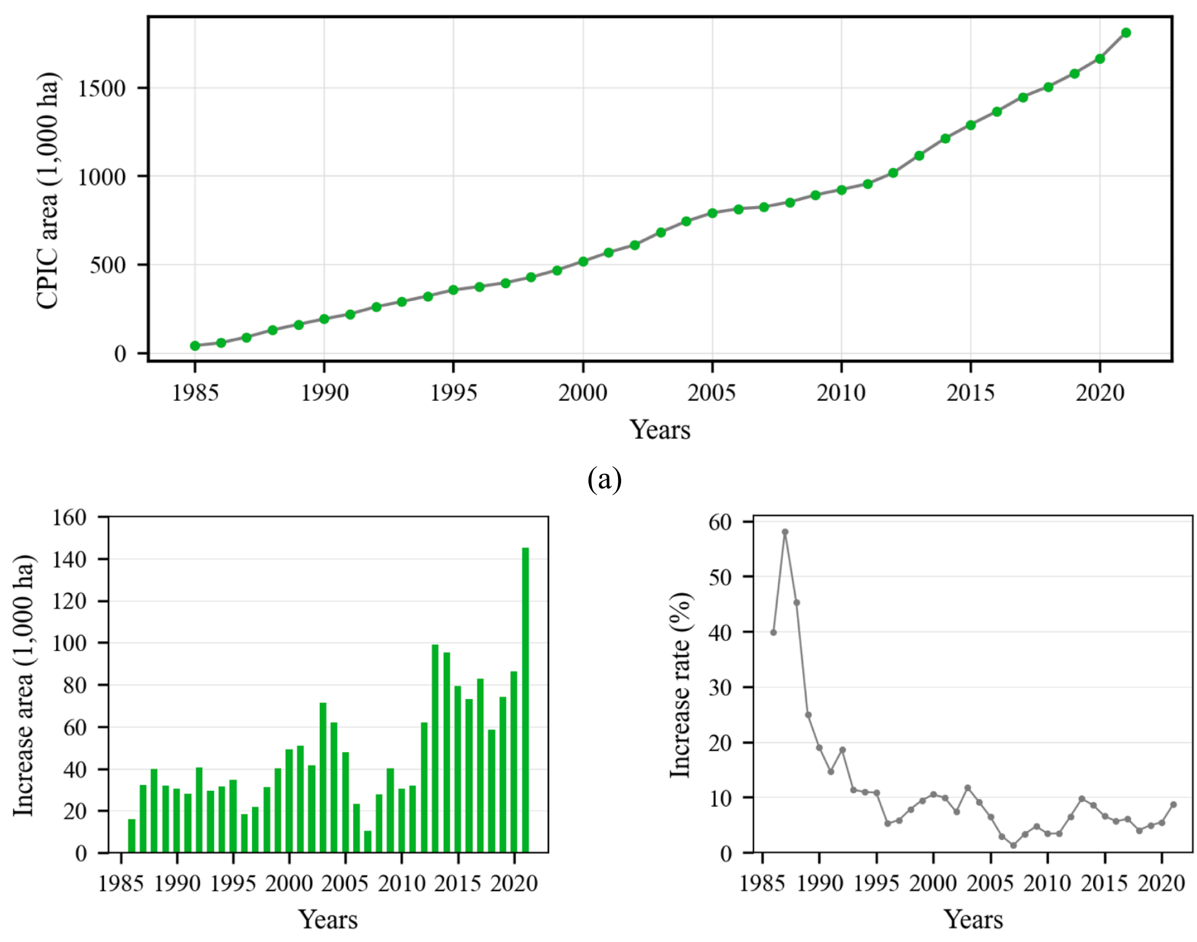

Fig. 14. Trend of the CPIC area from 1985 to 2021. (a) CPIC area from 1985 to 2021. (b) Annual increased area of CPIC from 1986 to 2021. (c) Annual increased rate of CPIC area from 1986 to 2021.

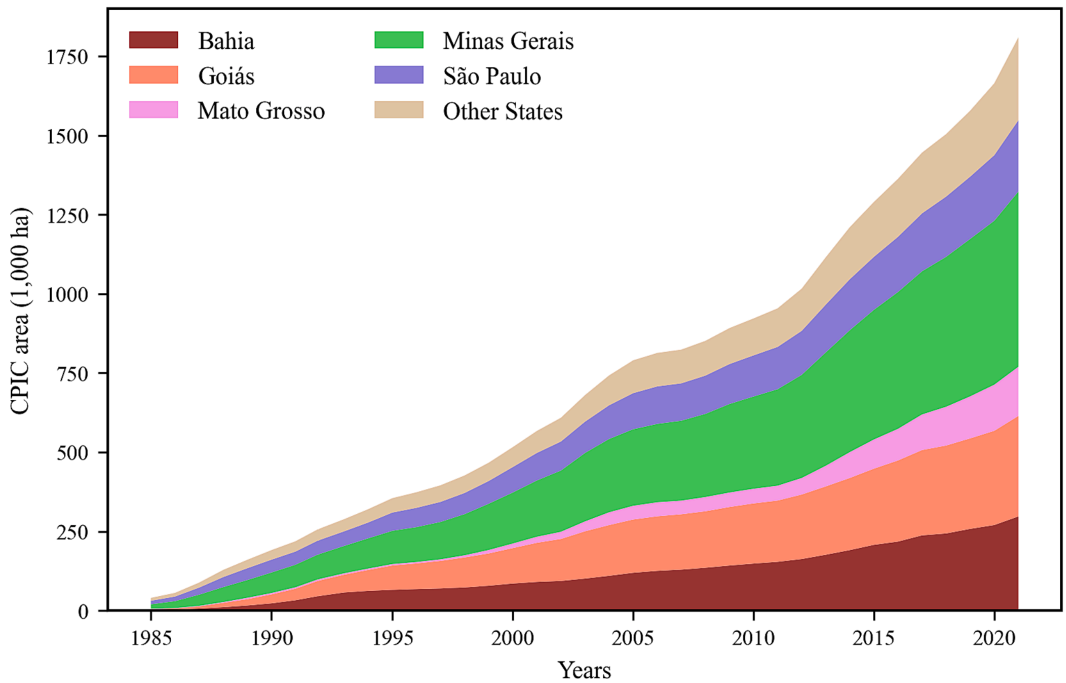

Fig. 15. Inter-annual variations of Center-Pivot Irrigated Cropland (CPIC) area in the top 5 states with the largest CPIC area expansions from 1985 to 2021.

this purpose, we constructed additional datasets involving different bands, which include 5, 7, 9, and 11 bands respectively. We trained U-Net models using these datasets separately and compared their prediction accuracy. Table 5 showed the prediction accuracy, the time cost in predicting images of size 6000 pixels × 6000 pixels, and the image storage required to complete the multi-year CPIC mapping of Brazil.

The results showed that adding new features can improve the accuracy of the model on the test dataset. In terms of the mIoU metric, the 9-band model has the highest accuracy, outperforming the 3-band model only by 0.005. Although the 9-band model achieved the highest accuracy, it was only 0.005 higher than the 3-band model on the mIoU metric. Moreover, the prediction time cost and image storage of the 9-band model were 2 and 3 times higher than those of the 3-band model, respectively.

The qualitative results showed that the models with more input bands had similar prediction results with the 3-band model (Fig. 18). Therefore, considering factors such as the time cost of prediction and data storage, this study used images composited with 3-band features to complete the multi-year CPIC mapping of Brazil.

We developed a data augmentation strategy named RandomMix that combines negative and positive samples to construct new samples, thereby increasing the proportion of CPIC pixels in the samples. We compared the difference in the accuracy of the models trained with or without the RandomMix strategy on our dataset, and the results were shown in Table 6. The results indicated that utilizing RandomMix can significantly improve the model's prediction accuracy, particularly in terms of recall, which improved from 0.828 to 0.888.

This result is consistent with our assumption. In the original dataset, the background pixels in the training samples accounted for a large proportion, which would bring data imbalance problems. This further leads to a serious bias in the classification model, which is more inclined to predict the sample as the background category, making the recall of the model usually low. The use of RandomMix increases the proportion of positive samples and mitigates the data imbalance problem, thus increasing the recall of the model and improving the prediction accuracy of the model.

5.2. ACM reliability and uncertainty

ACMs accurately captured Brazilian CPIC annual dynamics at the parcel-level spatial detail, representing the highest spatiotemporal resolution time-series data available for Brazil to date. The accuracy assessment on validation samples and ANA reported area demonstrated the reliability of our ACMs. The ANA data are also susceptible to errors, especially in earlier census years. For example, Fig. 19 visualized ANA's CPIC maps, our CPIC products, and the corresponding Landsat images in these years. The results illustrated that many CPIC areas were missed in ANA report maps. These misclassifications may be due to the scarcity of early data or the subjectivity of the interpretation experts. By contrast, our method detected all CPIC areas well, and our results had clear boundaries of CPIC and Non-CPIC areas, preserving the integrity of each CPIC (Fig. 19(c, f)). We can reasonably infer that in earlier census years (i.e., 1985 and 1990), Brazil had more CPICs than the ANA reported.

In addition, our ACMs are currently the only annual CPIC product that covers the entire Brazil and can support annual dynamic change analysis. ANA's data showed that CPIC has accelerated its growth since 2010 (ANA 2019). Our research further indicated that the time of accelerated expansion began in 2012, when the newly added area was 62,170 ha, twice that of 2011. In terms of spatial distribution, Minas Gerais, Goiás, Bahia, and São Paulo are still the states with the largest CPIC area in 2021, accounting for 30.6 %, 17.5 %, 16.4 %, and 12.4 % respectively. This proportion is consistent with the data from 2017 (ANA 2019).

Despite advances, there remain several factors that can affect the accuracy of our ACMs. The first uncertainty comes from the spatial gap of available Landsat image data due to cloud occlusion. The Landsat satellite has a 16-day revisit frequency and supports at most 23 effective observations in a year. Although using multiple Landsat sensors in combination can shorten the revisit period and increase the number of theoretical observations, the problem of insufficient observations is still encountered in the years with only a single sensor, which is particularly significant before 1999. Moreover, the number of effective observations is further reduced due to the influence of cloud pollution. Although we composite an image using the observational image of the whole year to maximize the spatial coverage of the image, some regions inevitably

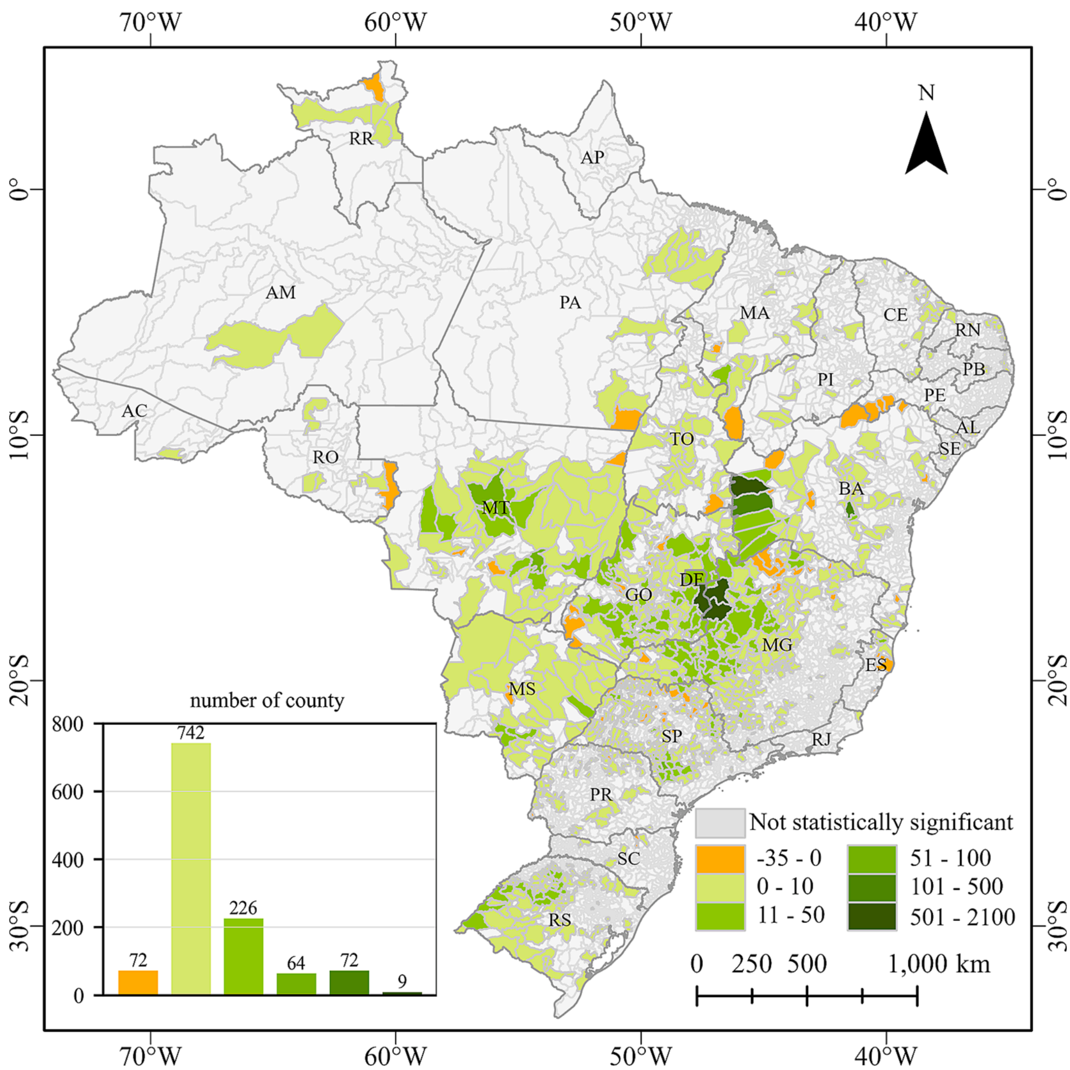

Fig. 16. Sen's slope of the CPIC area in each county from 1985 to 2021. Counties that do not pass the MK test are shown in gray.

remain unobservable for annual mapping at the national scale, which may lead to missed detection for CPIC areas.

The second uncertainty comes from the U-Net model and the spatial resolution of a Landsat image. The circular or fan-shaped CPIC in the image is the key feature of building the U-Net prediction model. When a CPIC occupies a few pixels, the circular shape will be lost, which may lead to the missed detection of the CPIC by the U-Net model. For example, some small CPIC areas with diameters of less than 150 m have appeared in recent years, and they only occupy 5 × 5 pixels in the Landsat image, which means presenting the circular shape of CPIC is difficult, as illustrated in Fig. 20.

Although small CPIC areas (less than 3 ha) only accounted for 0.06 % of the total area in 2019 (ANA 2019), they still need to introduce high-resolution data (e.g., Sentinel-2 image) to reduce this uncertainty in future works. In addition, the model uncertainty analysis methods (Gal & Ghahramani 2016; Kendall & Gal 2017) can provide prediction confidence pixel-by-pixel, which will help to identify areas of high uncertainty and optimize the existing segmentation methods in future work.

5.3. Expansion of CPIC encroaches on shrubs and forests

Forests in Brazil play a critical role in regulating carbon cycles and adjusting the global climate (Qin et al. 2019; Schlamadinger & Marland 1996). Many studies have indicated that cropland expansion in Brazil has been an essential driver of deforestation (Costa et al. 2007; Gibbs et al. 2010; Morton et al. 2006). CPIC has been one of the fastest-expanding croplands over the past few decades. Our ACMs revealed that the CPIC area in Brazil had expanded 45-fold since 1985 and is growing at an average rate of 100,000 ha per year. Therefore, determining which land types are occupied by newly increased CPIC areas, which is important for assessing their impacts on deforestation and making agricultural policies holistically, is urgent.

We collected land-cover products (Zhang et al. 2021) with a 30 m resolution from 1985 to 2020 (five-year interval) and explored the areas and proportions of different land-cover products converted to CPIC areas since 1985. The result showed that CPIC areas were mainly converted from herbaceous covers, shrublands, forests, rainfed croplands, and grasslands (Fig. 21).

Fig. 21 indicated that CPIC areas increased by 1.77 million ha from 1985, 50 % of which was converted from herbaceous covers, 24 % from shrublands, and 14 % from forests. That is, CPIC expansion destroyed 42,000 ha of shrubs and 25,000 ha of forests.

In addition, we exhibited the proportions and areas of croplands (including herbaceous covers and rainfed croplands), forests, shrublands, and grasslands that converted to CPIC areas in different periods, as illustrated in Fig. 22. The proportion results (Fig. 22a) showed that cropland was the main source of new CPIC areas, with an average contribution of about 81 %. The fraction of croplands converted to CPIC areas was rising, from 81 % to 89 %. By contrast, the proportions of forests and shrublands were declining, of which the fraction of forests

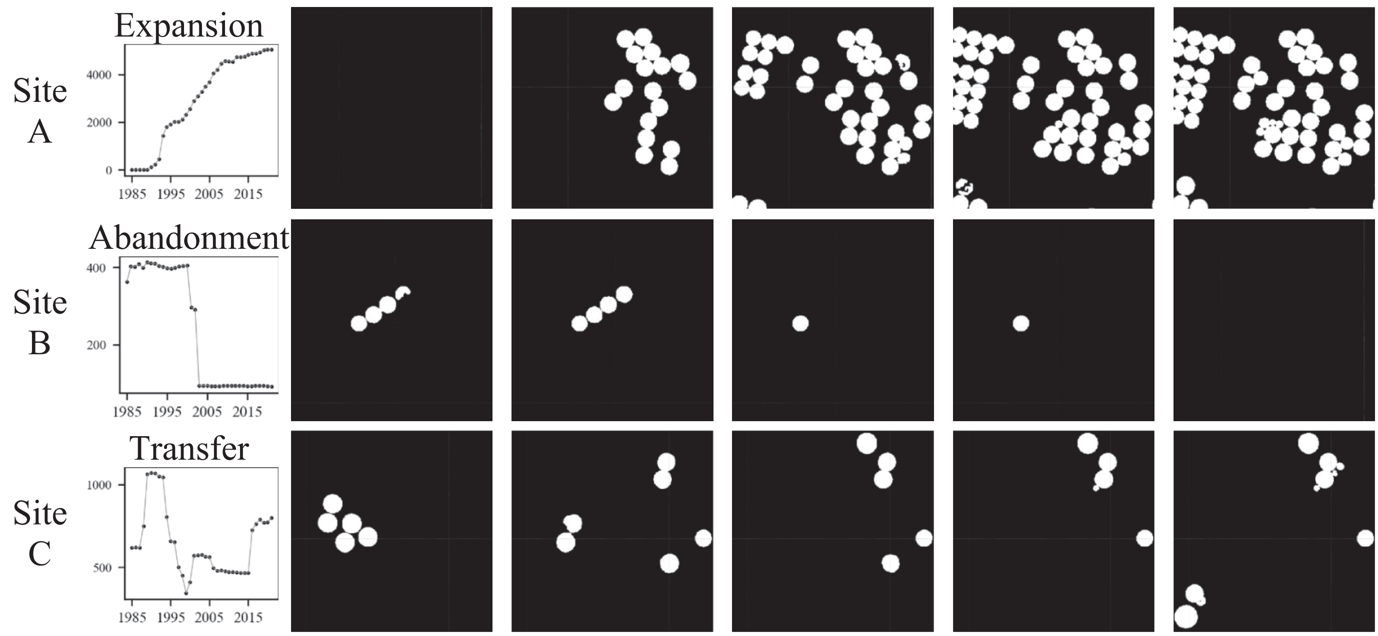

Fig. 17. CPIC change at pixel-level investigation from 1985 to 2021. Sites A, B, and C are located in Bahia, Pernambuco, and São Paulo, within which significant expansion, abandonment, and diversion changes have taken place, respectively. Annual CPIC area changes were captured in (a), and (b–f) show CPIC areas in the years 1985, 1995, 2005, 2015, and 2021.

Table 5
The comparison of models with different input bands.

Bands number	Composition of features	mIoU	Prediction time (s)	Storage (GB)
3	EVI_max + BSI_max + green	0.903	**6.8**	**2035**
5	+ NDMI_max + nir	0.907	9.9	3391
7	+ red + swir2	0.898	12.4	4748
9	+ NDVI_max + blue	**0.908**	17.3	6105
11	+ NDSMI_max + swir1	0.906	20.4	7461

decreased from 7 % to 3 % and that of shrublands decreased from 11 % to 7 %. The proportion of grassland changed very little, remaining at 1 %.

Despite the reduced fractions of forests and shrublands, the rapid CPIC expansion still led to the area increase of destroyed forests and shrubs (Fig. 22b). For example, the increase of CPIC areas between 2015 and 2020 resulted in the loss of 11,000 ha of forests and 25,000 ha of shrublands, the loss area is 1.3 times larger than that between 1985 and 1990. These results reflected that CPIC expansion was accelerating the encroachment of tree-growing regions, including forests and shrubs. Given that CPICs are still expanding rapidly, they may pose a non-negligible threat to Brazil's forest ecosystem.

Most new CPIC areas were converted from cropland, but in the long-term development process, this part of cropland may also be converted

Table 6
The effectiveness of the RandomMix on our datasets.

Model	RandomMix	mIoU	Precision	Recall	F1-score	OA
UNet	no	0.874	0.887	0.828	0.856	0.999
UNet	yes	0.903	0.898	0.888	0.893	0.999

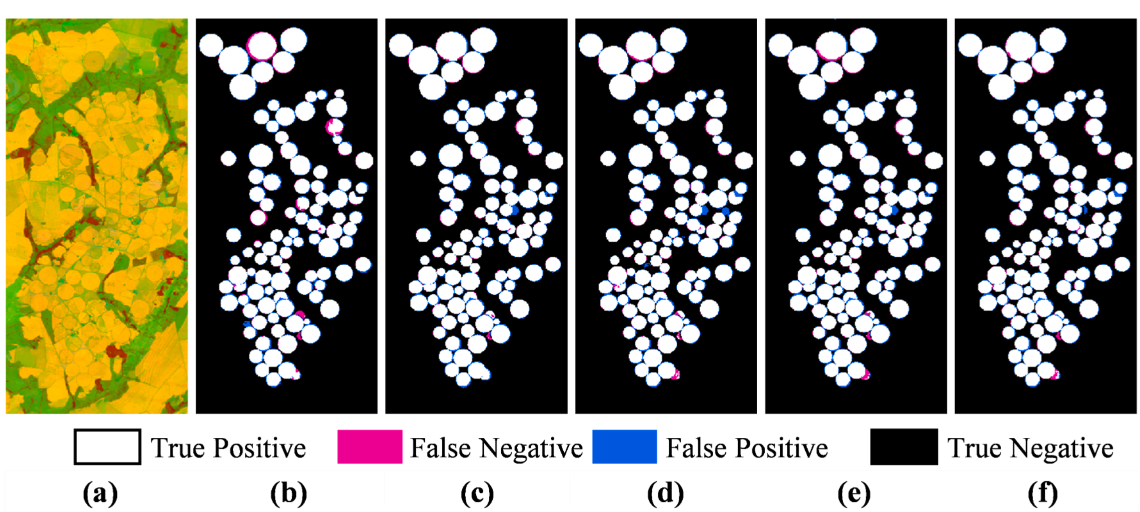

Fig. 18. The prediction results and errors of models using different input features on the test set. (a) Images of the test dataset. (b-f) are the models whose input features are 3, 5, 7, 9, and 11 bands respectively.

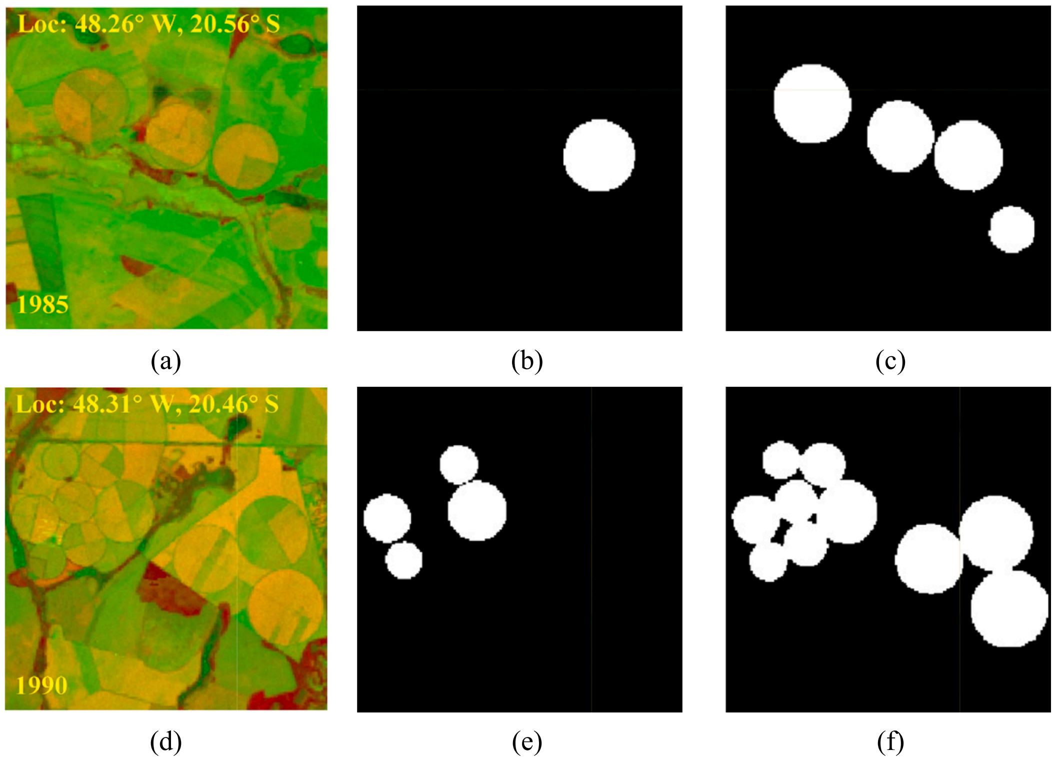

Fig. 19. Visualization of Landsat images (a, d), ANA report results (b, e), and our mapping results (c, f) for the years 1985 and 1990. The white and black in the results map indicate CPIC and non-CPIC, respectively.

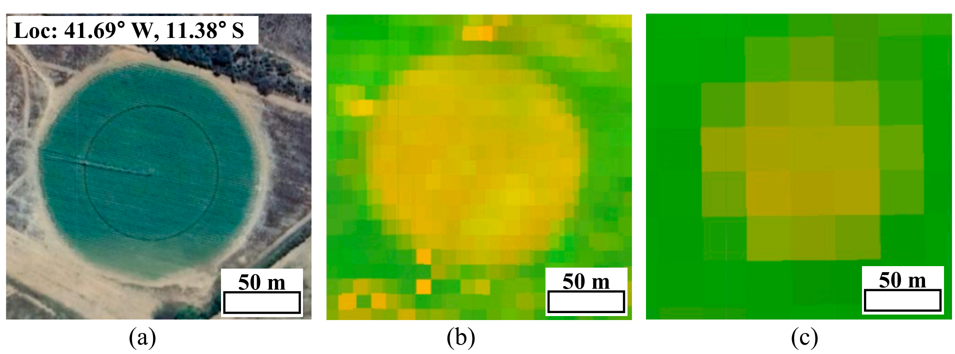

Fig. 20. A small CPIC (with a diameter of 150 m) presents different shapes in different spatial resolution images. (a) Google image with one-meter spatial resolution. (b) Sentinel-2 image with 10-meter spatial resolution. (c) Landsat image with 30-meter spatial resolution.

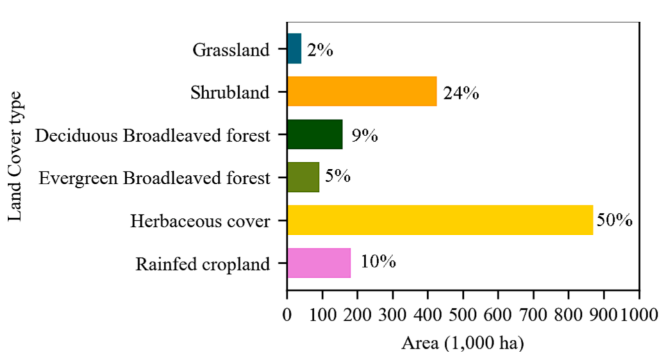

Fig. 21. Areas and proportions of different land-cover products converted to CPICs since 1985.

from other land-cover types. We traced the conversion process of land-cover types over the past few decades of increased CPIC regions in 2021 and demonstrated it with the Sankey diagram, as illustrated in Fig. 23.

The new CPIC in 2021 was 150,000 ha, 87 % of which was converted from cropland in 2020, 4 % from forest, and 7 % from shrubland. However, over the past few decades from 1985, shrubs and forests contributed 21 % and 17 %, respectively. Over time, shrubs and forests were progressively destroyed and converted into croplands, resulting in continuous cropland expansion (from 58 % to 87 %). Finally, croplands were converted to CPIC areas.

6. Conclusion

Mapping CPIC spatial distribution and its temporal dynamics over the long term is critical for understanding the expansion patterns and impacts on the CPIC ecosystem. We proposed a new framework to produce 30 m ACMs in Brazil from 1985 to 2021 by combining cloud

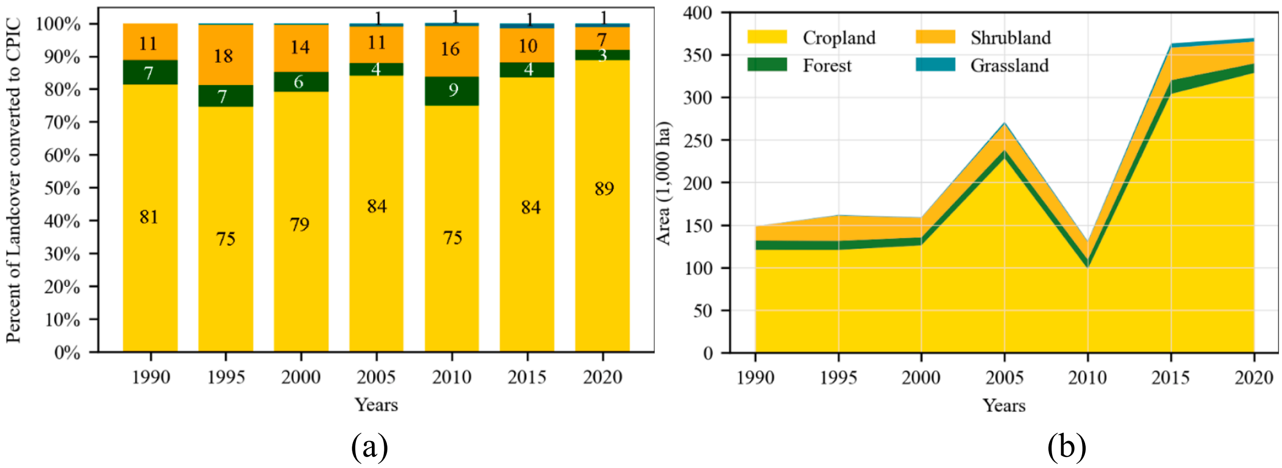

Fig. 22. Proportions (a) and areas (b) of croplands, forests, shrublands, and grasslands converted to CPIC areas in different periods.

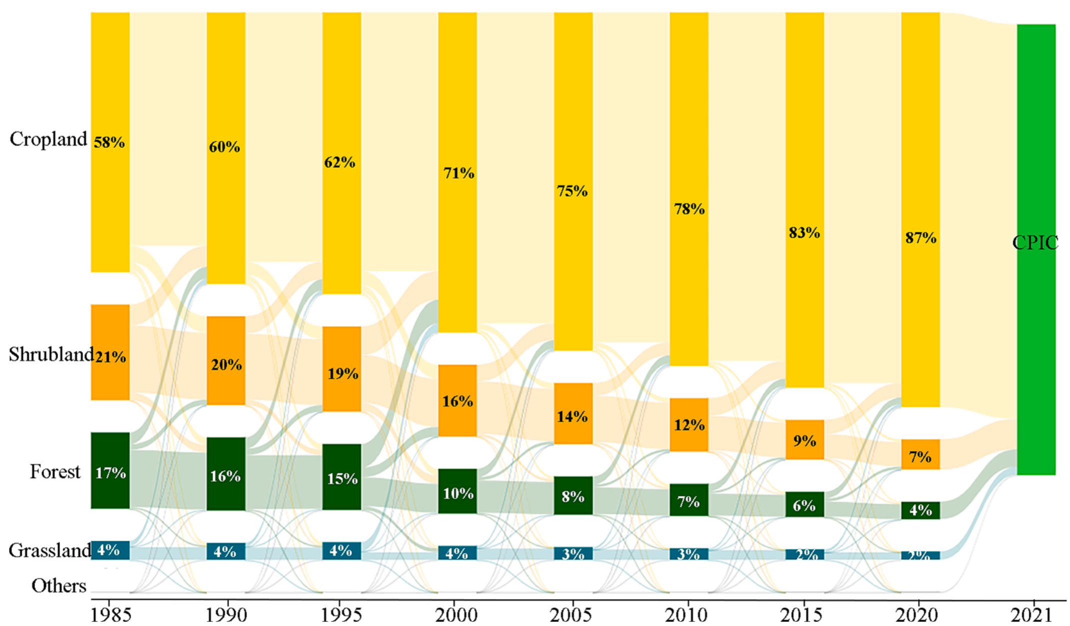

Fig. 23. Conversion process of land-cover products in the past decades in the increased CPIC area in 2021.

platforms and the U-Net model. The accuracy assessment demonstrated that ACMs had high quality and reliability with an average mIoU of 0.947 and the estimated area was highly consistent with official statistics. ACMs accurately captured the annual dynamics of Brazilian CPICs at the parcel-level spatial detail, representing the highest spatial and temporal resolution time-series products available for Brazil to date. Our results demonstrated that Brazilian CPICs expanded dramatically from 1985 to 2021, increasing by 45 times from 4×10^4 to 1.8×10^6 ha, and the expansion pace varied widely across states and periods. The CPIC expansion occurred mainly in the states of Minas Gerais, Goiás, and Bahia over the past decades, where increased CPIC areas exceeded 8,000 ha per year. Our analyses also indicated that CPIC expansion had encroached on 42,000 ha of shrubs and 25,000 ha of forests, and the impact was continuing and intensifying. Our framework and model can be directly applied to CPIC mapping in Brazil for future years without retraining. To the best of our knowledge, the produced ACMs are the longest and most spatially detailed CPIC products in Brazil that reflect the CPIC spatiotemporal changes in the last four decades. ACMs and analytical results contribute to informing relevant agricultural policy-making and environmental research.

CRediT authorship contribution statement

Xiangyu Liu: Conceptualization, Software, Methodology, Writing – original draft. **Wei He:** Writing – review & editing, Supervision, Validation. **Wenbin Liu:** Software, Visualization, Writing – review & editing. **Guoying Yin:** Writing – review & editing. **Hongyan Zhang:** Conceptualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.isprsjprs.2023.10.007.

References

ANA (2019). Levantamento da agricultura irrigada por pivôs centrais no Brasil (1985-2017). In: N.W.a.S. Agency (Ed.), pp. 9–10.

Alvares, C.A., Stape, J.L., Sentelhas, P.C., De Moraes Gonçalves, J.L., Sparovek, G., 2013. Köppen's climate classification map for Brazil. Meteorol. Zeitschrift 22, 711–728.

Belgiu, M., Csillik, O., 2018. Sentinel-2 cropland mapping using pixel-based and object-based time-weighted dynamic time warping analysis. Remote Sens. Environ. 204, 509–523.

Boulton, C.A., Lenton, T.M., Boers, N., 2022. Pronounced loss of Amazon rainforest resilience since the early 2000s. *Nature*. Clim. Change 12, 271.

Brandt, M., Tucker, C.J., Kariryaa, A., Rasmussen, K., Abel, C., Small, J., Chave, J., Rasmussen, L.V., Hiernaux, P., Diouf, A.A., Kergoat, L., Mertz, O., Igel, C., Gieseke, F., Schöning, J., Li, S., Melocik, K., Meyer, J., Sinno, S., Romero, E., Glennie, E., Montagu, A., Dendoncker, M., Fensholt, R., 2020. An unexpectedly large count of trees in the West African Sahara and Sahel. Nature 587, 78–82.

Breiman, L., 2001. Random forests. Mach. Learn. 45, 5–32.

Calderón-Loor, M., Hadjikakou, M., Bryan, B.A., 2021. High-resolution wall-to-wall land-cover mapping and land change assessment for Australia from 1985 to 2015. Remote Sens. Environ. 252, 112148.

Cardoso, D., Sarkinen, T., Alexander, S., Amorim, A.M., Bittrich, V., Celis, M., Daly, D.C., Fiaschi, P., Funk, V.A., Giacomin, L.L., Goldenberg, R., Heiden, G., Iganci, J., Kelloff, C.L., Knapp, S., de Lima, H.C., Machado, A.F.P., dos Santos, R.M., Mello-Silva, R., Michelangeli, F.A., Mitchell, J., Moonlight, P., de Moraes, P.L.R., Mori, S. A., Nunes, T.S., Pennington, T.D., Pirani, J.R., Prance, G.T., de Queiroz, L.P., Rapini, A., Riina, R., Rincon, C.A.V., Roque, N., Shimizu, G., Sobral, M., Stehmann, J.R., Stevens, W.D., Taylor, C.M., Trovo, M., van den Berg, C., van der Werff, H., Viana, P.L., Zartman, C.E., Forzza, R.C., 2017. Amazon plant diversity revealed by a taxonomically verified species list. PNAS 114, 10695–10700.

Carneiro, T., Da Nobrega, R.V.M., Nepomuceno, T., Bian, G.B., De Albuquerque, V.H.C., Filho, P.P.R., 2018. Performance Analysis of Google Colaboratory as a Tool for Accelerating Deep Learning Applications. IEEE Access 6, 61677–61685.

Chai, Q., Gan, Y.T., Zhao, C., Xu, H.L., Waskom, R.M., Niu, Y.N., Siddique, K.H.M., 2016. Regulated deficit irrigation for crop production under drought stress. A review. Agron. Sustain. Dev. 36.

Cooley, D., Smith, S., 2022. Center Pivot Irrigation Systems as a Form of Drought Risk Mitigation in Humid Regions. National Bureau of Economic Research.

Costa, M.H., Yanagi, S.N.M., Souza, P.J.O.P., Ribeiro, A., Rocha, E.J.P., 2007. Climate change in Amazonia caused by soybean cropland expansion, as compared to caused by pastureland expansion. Geophys. Res. Lett. 34.

de Albuquerque, A.O., de Carvalho, O.A., de Carvalho, O.L.F., de Bem, P.P., Guimarães, P.H., Ferreira, R., dos Santos de Moura, Silva, C.R., Trancoso Gomes, R. A., Guimarães, R.F., 2020. Deep semantic segmentation of center pivot irrigation systems from remotely sensed data. Remote Sens. (Basel) 12.

de Albuquerque, A.O., de Carvalho, O.L.F., e Silva, C.R., de Bem, P.P., Gomes, R.A.T., Borges, D.L., Guimarães, R.F., Pimentel, C.M.M., Carvalho Júnior, C.A., 2021a. Instance segmentation of center pivot irrigation systems using multi-temporal SENTINEL-1 SAR images. Rem. Sens. Applicat. Soc. Environ. 23, 100537.

de Albuquerque, A.O., de Carvalho, O.L.F., e Silva, C.R., Luiz, A.S., de Bem, P.P., Gomes, R.A.T., Guimarães, R.F., de Carvalho Júnior, O.A., 2021b. Dealing with clouds and seasonal changes for center pivot irrigation systems detection using instance segmentation in sentinel-2 time series. IEEE J. Select. Top. Appl. Earth Observ. Rem. Sens. 14, 8447–8457.

de Carvalho, O.L.F., de Carvalho, O.A., Albuquerque, A.O.D., Bem, P.P.D., Silva, C.R., Ferreira, P.H.G., de Moura, R.D.S., Gomes, R.A.T., Guimarães, R.F., Borges, D.L., 2021. Instance segmentation for large, multi-channel remote sensing imagery using mask-RCNN and a mosaicking approach. Rem. Sens. 13, 1–24.

Deines, J.M., Kendall, A.D., Hyndman, D.W., 2017. Annual irrigation dynamics in the U. S. Northern high plains derived from landsat satellite data. Geophys. Res. Lett. 44, 9350–9360.

Descals, A., Wich, S., Meijaard, E., Gaveau, D.L.A., Peedell, S., Szantoi, Z., 2021. High-resolution global map of smallholder and industrial closed-canopy oil palm plantations. Earth Syst. Sci. Data 13, 1211–1231.

Diek, S., Fornallaz, F., Schaepman, M.E., de Jong, R., 2017. Barest Pixel Composite for agricultural areas using landsat time series. Remote Sens. (Basel) 9.

Duda, R.O., Hart, P.E., 1972. Use of the Hough transformation to detect lines and curves in pictures. Commun. ACM 15, 11–15.

FAO (2017). The future of food and agriculture: Trends and challenges. In. Rome.

FAO (2020). World Food and Agriculture - Statistical Yearbook 2020. In. Rome.

Gal, Y., & Ghahramani, Z. (2016). Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In: *International conference on machine learning*, PMLR, pp. 1050-1059.

Gao, B.-C., 1996. NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space. Remote Sens. Environ. 58, 257–266.

Gibbs, H.K., Ruesch, A.S., Achard, F., Clayton, M.K., Holmgren, P., Ramankutty, N., & Foley, J.A. (2010). Tropical forests were the primary sources of new agricultural land in the 1980s and 1990s. In: *Proceedings of the National Academy of Sciences of the United States of America, 107*, pp. 16732-16737.

Gong, P., Li, X., Wang, J., Bai, Y., Chen, B., Hu, T., Liu, X., Xu, B., Yang, J., Zhang, W., 2020. Annual maps of global artificial impervious area (GAIA) between 1985 and 2018. Remote Sens. Environ. 236, 111510.

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., Moore, R., 2017. Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sens. Environ. 202, 18–27.

Hansen, M.C., Potapov, P.V., Moore, R., Hancher, M., Turubanova, S.A., Tyukavina, A., Thau, D., Stehman, S.V., Goetz, S.J., Loveland, T.R., Kommareddy, A., Egorov, A., Chini, L., Justice, C.O., Townshend, J.R.G., 2013. High-resolution global maps of 21st-century forest cover change. Science 342, 850–853.

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In: *Proceedings of the IEEE conference on computer vision and pattern recognition,* pp. 770-778.

Huete, A.R., Liu, H.Q., Batchily, K., Van Leeuwen, W., 1997. A comparison of vegetation indices over a global set of TM images for EOS-MODIS. Remote Sens. Environ. 59, 440–451.

Johansen, K., Lopez, O., Tu, Y.H., Li, T., McCabe, M.F., 2021. Center pivot field delineation and mapping: A satellite-driven object-based image analysis approach for national scale accounting. ISPRS J. Photogramm. Remote Sens. 175, 1–19.

Kang, S.Z., Hao, X.M., Du, T.S., Tong, L., Su, X.L., Lu, H.N., Li, X.L., Huo, Z.L., Li, S.E., Ding, R.S., 2017. Improving agricultural water productivity to ensure food security in China under changing environment: From research to practice. Agric Water Manag 179, 5–17.

Kattenborn, T., Leitloff, J., Schiefer, F., Hinz, S., 2021. Review on Convolutional Neural Networks (CNN) in vegetation remote sensing. ISPRS J. Photogramm. Remote Sens. 173, 24–49.

Kendall, A., & Gal, Y. (2017). What uncertainties do we need in bayesian deep learning for computer vision? *Adv. Neural Inform. Processing Syst.* 30.

Kendall, M.G. (1946). The advanced theory of statistics. *The advanced theory of statistics. 2nd ed.*

Kruitwagen, L., Story, K.T., Friedrich, J., Byers, L., Skillman, S., Hepburn, C., 2021. A global inventory of photovoltaic solar energy generating units. Nature 598, 604–610.

LeCun, Y., Bengio, Y., Hinton, G., 2015. Deep learning. Nature 521, 436–444.

Li, X., Gong, P., Liang, L., 2015. A 30-year (1984–2013) record of annual urban dynamics of Beijing City derived from Landsat data. Remote Sens. Environ. 166, 78–90.

Li, S., Song, W., Qin, H., Hao, A., 2018. Deep variance network: An iterative, improved CNN framework for unbalanced training datasets. Pattern Recogn. 81, 294–308.

Li, X., Yu, L., Peng, D., Gong, P., 2021. A large-scale, long time-series (1984–2020) of soybean mapping with phenological features: Heilongjiang Province as a test case. Int. J. Remote Sens. 42, 7332–7356.

Loveland, T.R., Dwyer, J.L., 2012. Landsat: Building a strong future. Remote Sens. Environ. 122, 22–29.

Luo, Y.C., Zhang, Z., Chen, Y., Li, Z.Y., Tao, F.L., 2020. ChinaCropPhen1km: a high-resolution crop phenological dataset for three staple crops in China during 2000–2015 based on leaf area index (LAI) products. Earth Syst. Sci. Data 12, 197–214.

Mann, H.B., 1945. Nonparametric tests against trend. Econometrica 13, 245–259.

McConnell, W.J., Viña, A., 2018. Interactions between food security and land use in the context of global change. Land 7, 53.

Morton, D.C., DeFries, R.S., Shimabukuro, Y.E., Anderson, L.O., Arai, E., Del Bon Espirito-Santo, F., Freitas, R., Morisette, J., 2006. Cropland expansion changes deforestation dynamics in the southern Brazilian Amazon. PNAS 103, 14637–14641.

Multsch, S., Krol, M.S., Pahlow, M., Assunção, A.L.C., Barretto, A.G.O.P., De Jong Van Lier, Q., Breuer, L., 2020. Assessment of potential implications of agricultural irrigation policy on surface water scarcity in Brazil. Hydrol. Earth Syst. Sci. 24, 307–324.

Olofsson, P., Foody, G.M., Stehman, S.V., Woodcock, C.E., 2013. Making better use of accuracy data in land change studies: Estimating accuracy and area and quantifying uncertainty using stratified estimation. Remote Sens. Environ. 129, 122–131.

Pekel, J.F., Cottam, A., Gorelick, N., Belward, A.S., 2016. High-resolution mapping of global surface water and its long-term changes. Nature 540, 418–422.

Qin, Y., Xiao, X., Dong, J., Zhang, Y., Wu, X., Shimabukuro, Y., Arai, E., Biradar, C., Wang, J., Zou, Z., Liu, F., Shi, Z., Doughty, R., Moore III, B., 2019. Improved estimates of forest cover and loss in the Brazilian Amazon in 2000–2017. Nat. Sustainability 2, 764–772.

Rodrigues, M.L., Korting, T.S., de Queiroz, G.R., Sales, C.P., da Silva, L.A.R., & Ieee (2020). Detecting Center Pivots in Matopiba using Hough Transform and Web Time Series Service. In: *IEEE Latin American GRSS and ISPRS Remote Sensing Conference (LAGIRS)*. Santiago, CHILE, pp. 189-194.

Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 234–241.

Rosa, L., Chiarelli, D.D., Rulli, M.C., Dell'Angelo, J., D'Odorico, P., 2020. Global agricultural economic water scarcity. *Science*. Advances 6, eaaz6031.

Rulli, M.C., Saviori, A., D'Odorico, P., 2013. Global land and water grabbing. PNAS 110, 892–897.

Salehi, S.S.M., Erdogmus, D., Gholipour, A., 2017. Tversky loss function for image segmentation using 3D fully convolutional deep networks. International Workshop on Machine Learning in Medical Imaging.

Saraiva, M., Protas, É., Salgado, M., Souza, C., 2020. Automatic mapping of center pivot irrigation systems from satellite images using deep learning. Remote Sens. (Basel) 12, 1–14.

Schlamadinger, B., Marland, G., 1996. The role of forest and bioenergy strategies in the global carbon cycle. Biomass Bioenergy 10, 275–300.

Sen, P.K., 1968. Estimates of the Regression Coefficient Based on Kendall's Tau. J. Am. Stat. Assoc. 63, 1379–1389.

Silverio, D., Brando, P.M., Macedo, M.N., Beck, P.S.A., Bustamante, M., Coe, M.T., 2015. Agricultural expansion dominates climate changes in southeastern Amazonia: the overlooked non-GHG forcing. Environ. Res. Lett. 10.

Sirko, W., Kashubin, S., Ritter, M., Annkah, A., Bouchareb, Y.S.E., Dauphin, Y., Keysers, D., Neumann, M., Cisse, M., Quinn, J., 2021. Continental-Scale Building Detection from High Resolution Satellite Imagery. arXiv:2107.12283 1–15.

Spera, S.A., Galford, G.L., Coe, M.T., Macedo, M.N., Mustard, J.F., 2016. Land-use change affects water recycling in Brazil's last agricultural frontier. Glob. Chang. Biol. 22, 3405–3413.

Sun, K., Xiao, B., Liu, D., Wang, J., 2019. Deep high-resolution representation learning for human pose estimation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 5693–5703.

Tan, M., Le, Q., 2019. In: Efficientnet: Rethinking Model Scaling for Convolutional Neural Networks. PMLR, pp. 6105–6114.

Tang, J., Arvor, D., Corpetti, T., Tang, P., 2021a. Mapping center pivot irrigation systems in the southern amazon from sentinel-2 images. Water 13, 1–17.

Tang, J., Zhang, Z., Zhao, L., Tang, P., 2021b. Increasing shape bias to improve the precision of center pivot irrigation system detection. Remote Sens. (Basel) 13, 612.

Tucker, C.J., 1979. Red and photographic infrared linear combinations for monitoring vegetation. Remote Sens. Environ. 8, 127–150.

Wang, X.H., Muller, C., Elliot, J., Mueller, N.D., Ciais, P., Jagermeyr, J., Gerber, J., Dumas, P., Wang, C.Z., Yang, H., Li, L., Deryng, D., Folberth, C., Liu, W.F., Makowski, D., Olin, S., Pugh, T.A.M., Reddy, A., Schmid, E., Jeong, S., Zhou, F., Piao, S., 2021. Global irrigation contribution to wheat and maize yield. Nat. Commun. 12, 1235.

Wulder, M.A., Roy, D.P., Radeloff, V.C., Loveland, T.R., Anderson, M.C., Johnson, D.M., Healey, S., Zhu, Z., Scambos, T.A., Pahlevan, N., Hansen, M., Gorelick, N., Crawford, C.J., Masek, J.G., Hermosilla, T., White, J.C., Belward, A.S., Schaaf, C., Woodcock, C. E., Huntington, J.L., Lymburner, L., Hostert, P., Gao, F., Lyapustin, A., Pekel, J.-F., Strobl, P., & Cook, B.D. (2022). Fifty years of Landsat science and impacts. *Rem. Sens. Environ. 280*.

Wulder, M.A., Loveland, T.R., Roy, D.P., Crawford, C.J., Masek, J.G., Woodcock, C.E., Allen, R.G., Anderson, M.C., Belward, A.S., Cohen, W.B., Dwyer, J., Erb, A., Gao, F., Griffiths, P., Helder, D., Hermosilla, T., Hipple, J.D., Hostert, P., Hughes, M.J., Huntington, J., Johnson, D.M., Kennedy, R., Kilic, A., Li, Z., Lymburner, L., McCorkel, J., Pahlevan, N., Scambos, T.A., Schaaf, C., Schott, J.R., Sheng, Y., Storey, J., Vermote, E., Vogelmann, J., White, J.C., Wynne, R.H., Zhu, Z., 2019. Current status of Landsat program, science, and applications. Remote Sens. Environ. 225, 127–147.

Xie, Y., Lark, T.J., Brown, J.F., Gibbs, H.K., 2019a. Mapping irrigated cropland extent across the conterminous United States at 30 m resolution using a semi-automatic training approach on Google Earth Engine. ISPRS J. Photogramm. Remote Sens. 155, 136–149.

Xie, Z., Phinn, S.R., Game, E.T., Pannell, D.J., Hobbs, R.J., Briggs, P.R., McDonald-Madden, E., 2019b. Using Landsat observations (1988–2017) and Google Earth Engine to detect vegetation cover changes in rangelands - A first step towards identifying degraded lands for conservation. Remote Sens. Environ. 232, 111317.

Xiong, J., Thenkabail, P.S., Gumma, M.K., Teluguntla, P., Poehnelt, J., Congalton, R.G., Yadav, K., Thau, D., 2017. Automated cropland mapping of continental Africa using Google Earth Engine cloud computing. ISPRS J. Photogramm. Remote Sens. 126, 225–244.

Yang, J., Huang, X., 2021. The 30 m annual land cover dataset and its dynamics in China from 1990 to 2019. Earth Syst. Sci. Data 13, 3907–3925.

Yue, J., Tian, J., Tian, Q., Xu, K., Xu, N., 2019. Development of soil moisture indices from differences in water absorption between shortwave-infrared bands. ISPRS J. Photogramm. Remote Sens. 154, 216–230.

Yun, S., Han, D., Oh, S.J., Chun, S., Choe, J., Yoo, Y., & Ieee (2019). CutMix: Regularization strategy to train strong classifiers with localizable features. In: *IEEE/CVF International Conference on Computer Vision (ICCV)*, Seoul, SOUTH KOREA, pp. 6022-6031.

Zhang, X., Liu, L., Chen, X., Gao, Y., Xie, S., Mi, J., 2021. GLC_FCS30: global land-cover product with fine classification system at 30m using time-series Landsat imagery. Earth Syst. Sci. Data 13, 2753–2776.

Zhang, H., Liu, W., Zhang, L., 2022. Seamless and automated rapeseed mapping for large cloudy regions using time-series optical satellite imagery. ISPRS J. Photogramm. Remote Sens. 184, 45–62.

Zhang, C., Zhang, H., Tian, S., 2023. Phenology-assisted supervised paddy rice mapping with the Landsat imagery on Google Earth Engine: Experiments in Heilongjiang Province of China from 1990 to 2020. Comput. Electron. Agric. 212, 108105.

Zhu, Z., Woodcock, C.E., 2014. Automated cloud, cloud shadow, and snow detection in multitemporal Landsat data: An algorithm designed specifically for monitoring land cover change. Remote Sens. Environ. 152, 217–234.